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(54) **DIGITAL PRODUCT MANAGEMENT SUPPORT METHOD, DIGITAL PRODUCT MANAGEMENT SUPPORT DEVICE, DIGITAL PRODUCT MANAGEMENT SUPPORT PROGRAM, AND DIGITAL PRODUCT MANAGEMENT SUPPORT SYSTEM**

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CPC ***G06Q 10/103*** (2013.01); ***G06F 16/2379*** (2019.01); ***G06Q 20/06*** (2013.01); ***G06Q 20/42*** (2013.01)

(57) **ABSTRACT**

A dealing terminal requests a distributed ledger system to register a transaction including owner information in which a purchaser of a digital product is an owner and a fact that a seller of the digital product has a right to claim payment for the digital product. When a deposit notification of a purchase amount of the digital product is received, the dealing terminal requests the distributed ledger system to register a transaction including the fact that the seller of the digital product does not have the right to claim payment for the digital product. When an actual product is sold, the dealing terminal requests the distributed ledger system to register a transaction including the fact that the purchaser of the actual product is the owner of the digital product and a location of the actual product is registered as a location of the digital product.

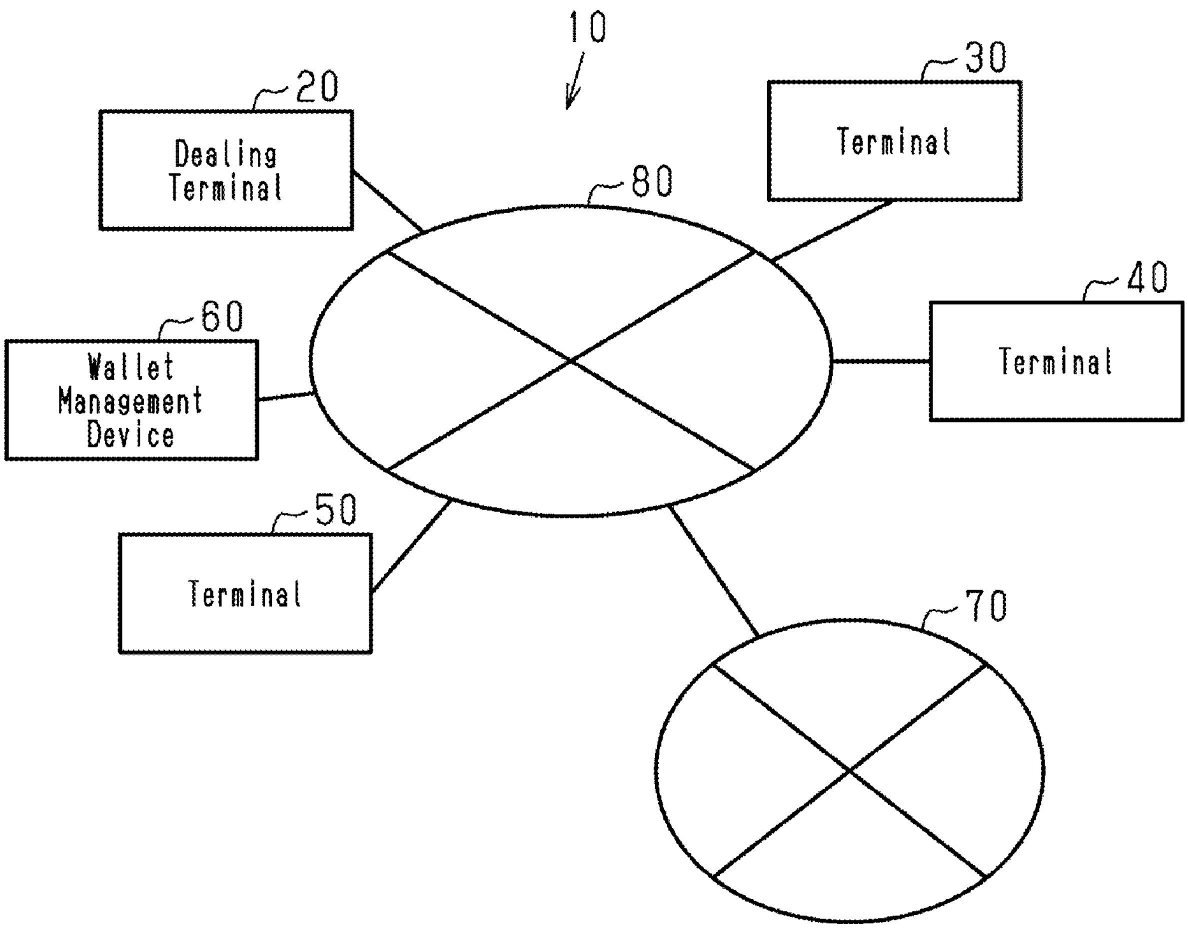


Fig.1

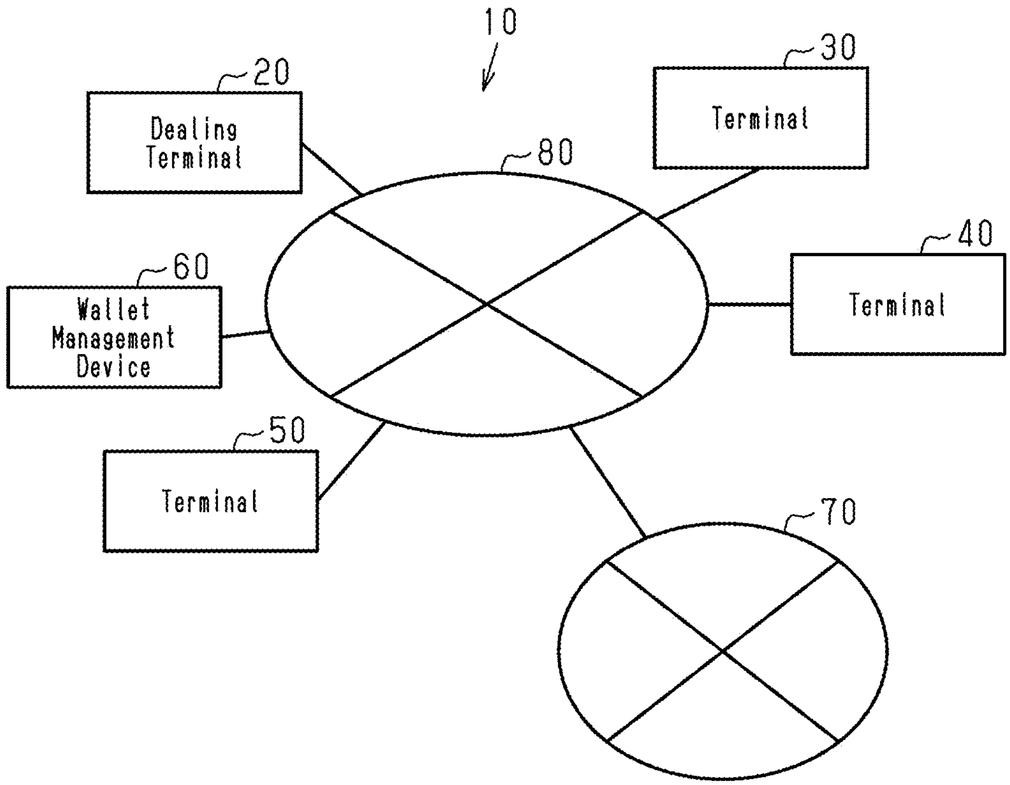

10
20
Dealing Terminal
30
Terminal
80
40
Terminal
60
Wallet Management Device
50
Terminal
70

Fig.2

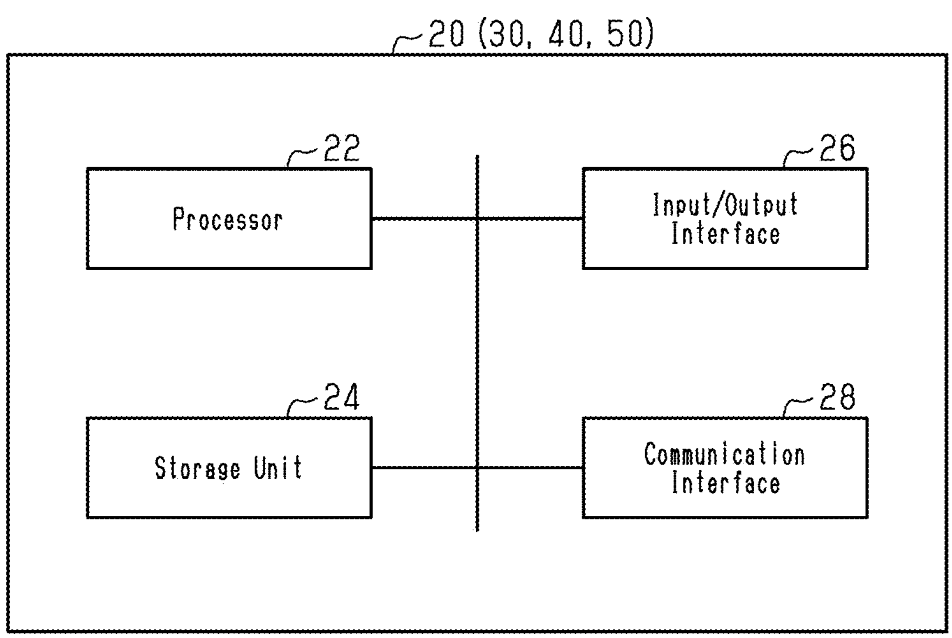

20 (30, 40, 50)
22
Processor
26
Input/Output Interface
24
Storage Unit
28
Communication Interface

Fig.3

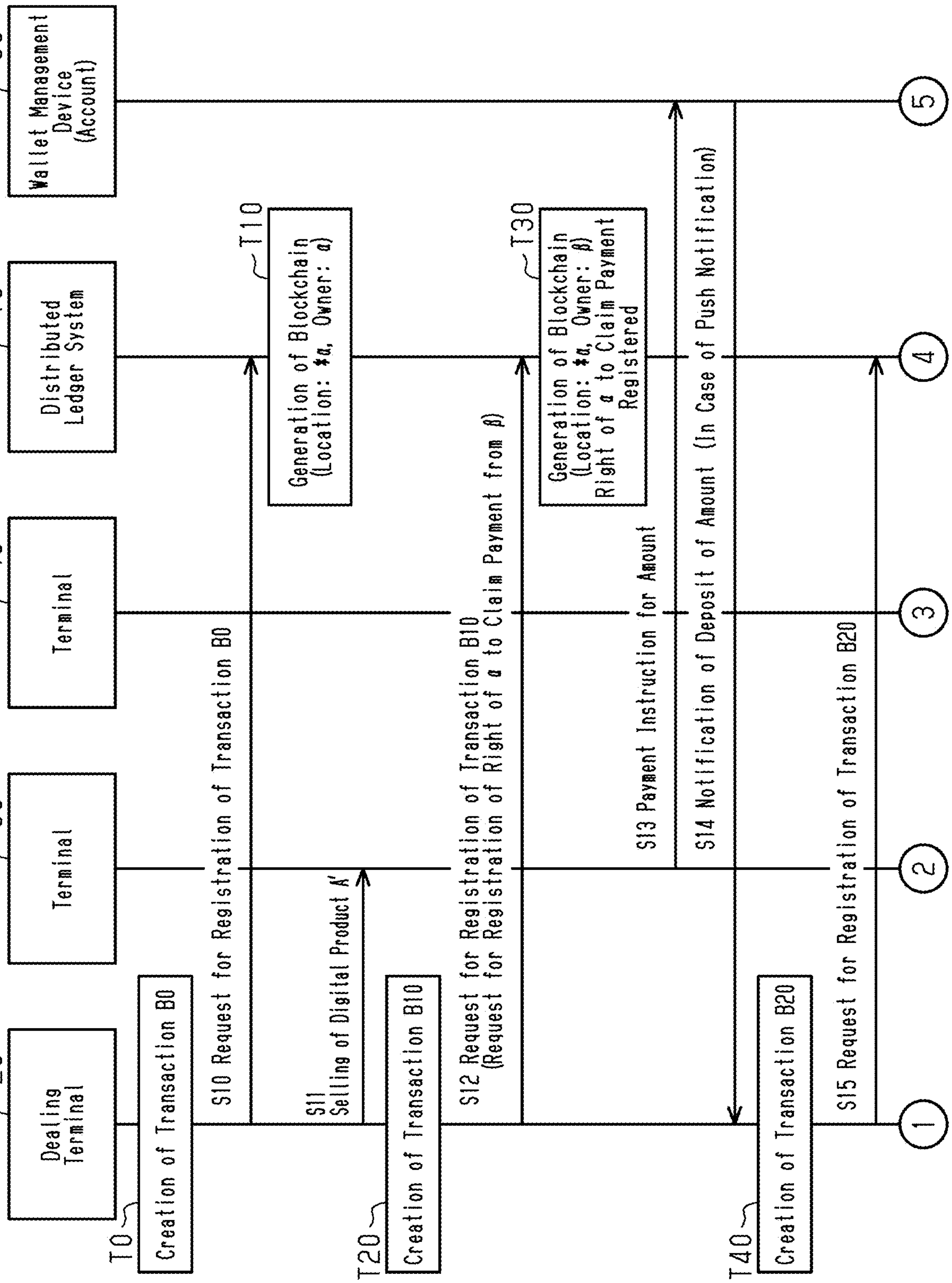

20 Dealing Terminal
30 Terminal
40 Terminal
70 Distributed Ledger System
60 Wallet Management Device (Account)
T0 Creation of Transaction B0
S10 Request for Registration of Transaction B0
T10 Generation of Blockchain (Location: #α, Owner: α)
S11 Selling of Digital Product A'
T20 Creation of Transaction B10
S12 Request for Registration of Transaction B10 (Request for Registration of Right of α to Claim Payment from β)
T30 Generation of Blockchain (Location: #α, Owner: β) Right of α to Claim Payment Registered
S13 Payment Instruction for Amount
S14 Notification of Deposit of Amount (In Case of Push Notification)
T40 Creation of Transaction B20
S15 Request for Registration of Transaction B20
1
2
3
4
5

Fig.4

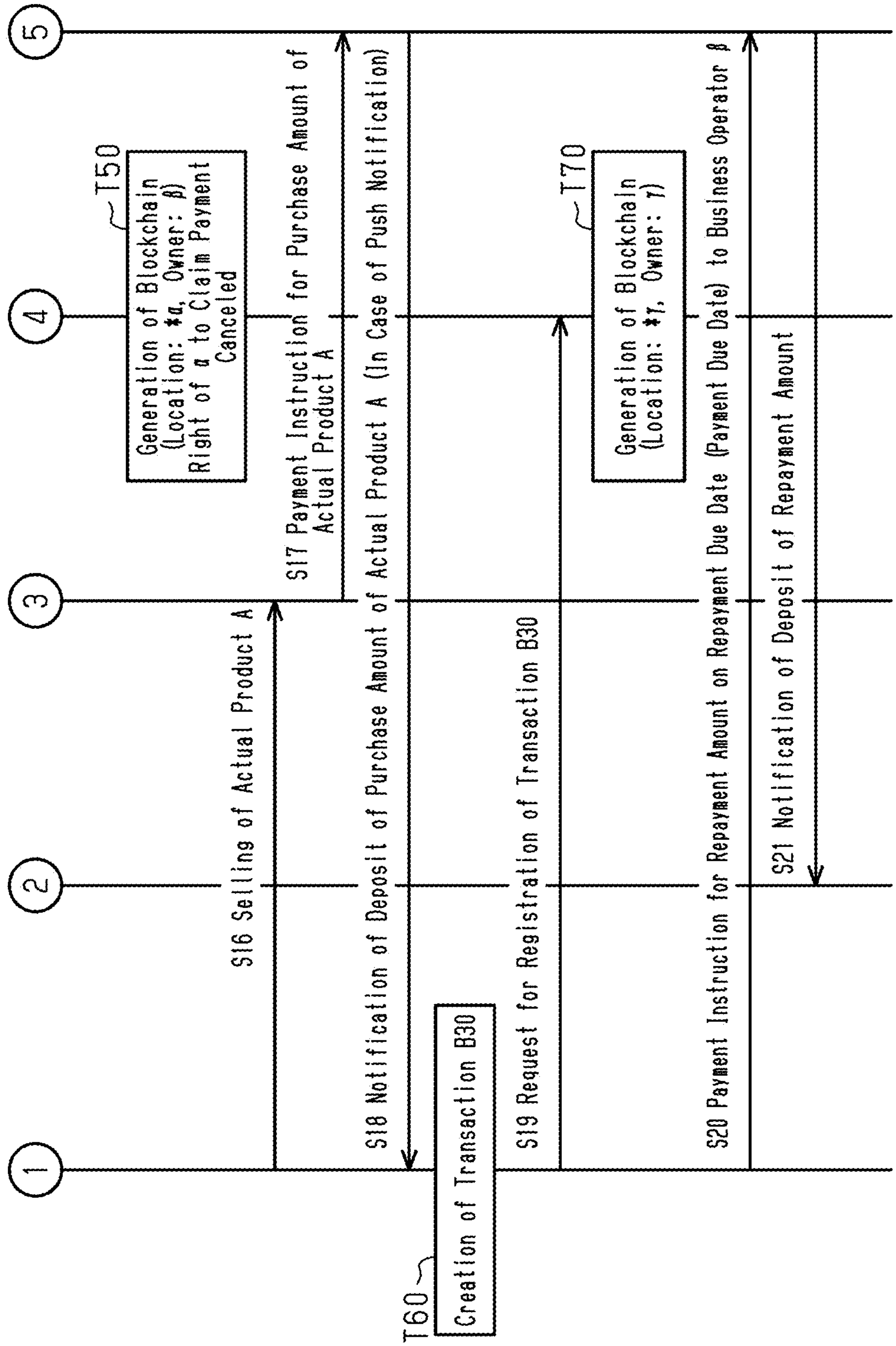
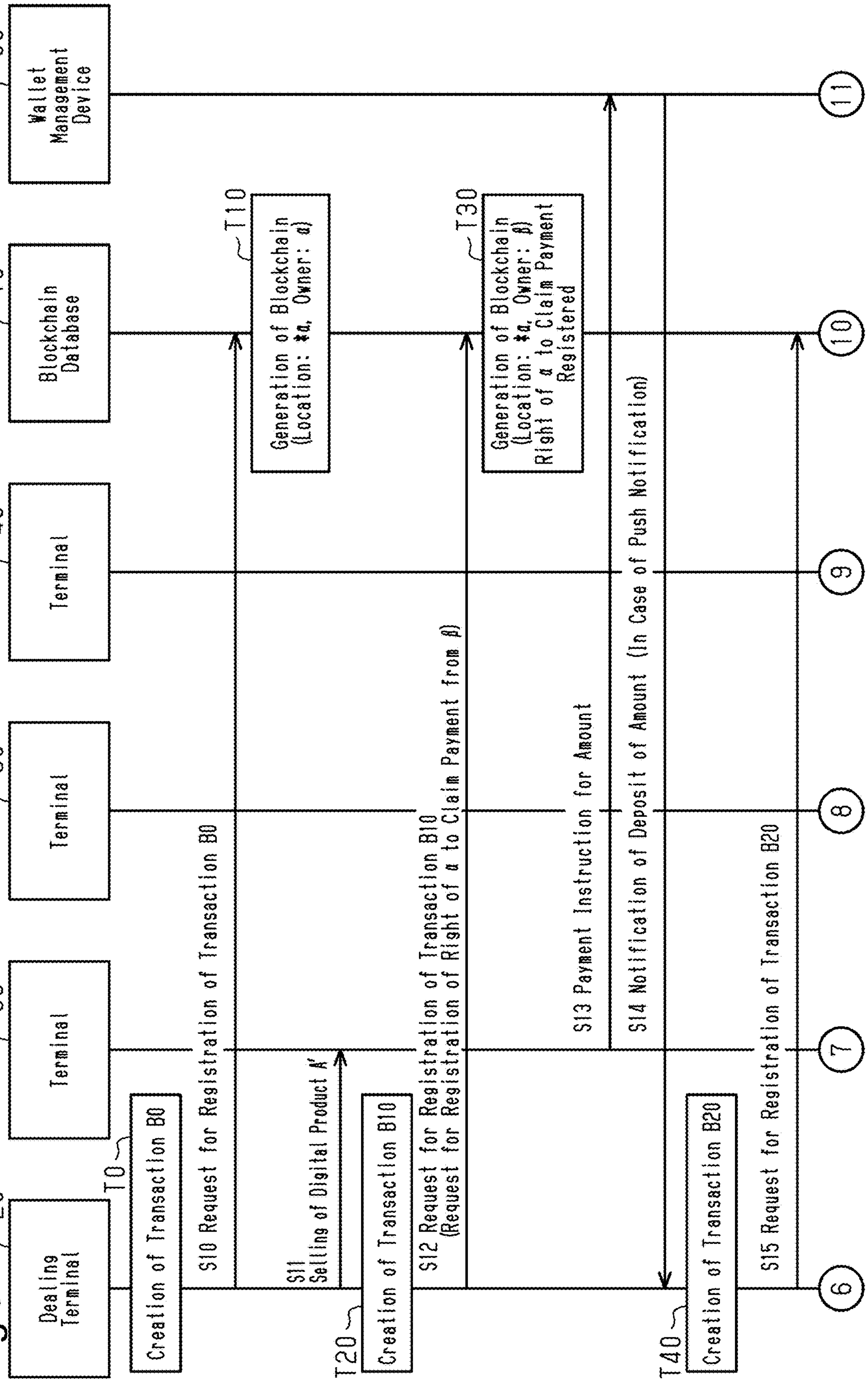

1
2
3
4
5
T50
Generation of Blockchain (Location: *α, Owner: β) Right of α to Claim Payment Canceled
T70
Generation of Blockchain (Location: *γ, Owner: γ)
T60
Creation of Transaction B30
S16 Selling of Actual Product A
S17 Payment Instruction for Purchase Amount of Actual Product A
S18 Notification of Deposit of Purchase Amount of Actual Product A (In Case of Push Notification)
S19 Request for Registration of Transaction B30
S20 Payment Instruction for Repayment Amount on Repayment Due Date (Payment Due Date) to Business Operator β
S21 Notification of Deposit of Repayment Amount

Fig.5



20 Dealing Terminal
30 Terminal
50 Terminal
40 Terminal
70 Blockchain Database
60 Wallet Management Device
T0 Creation of Transaction B0
S10 Request for Registration of Transaction B0
T10 Generation of Blockchain (Location: ¥α, Owner: α)
S11 Selling of Digital Product A'
T20 Creation of Transaction B10
S12 Request for Registration of Transaction B10 (Request for Registration of Right of α to Claim Payment from β)
T30 Generation of Blockchain (Location: ¥α, Owner: β) Right of α to Claim Payment Registered
S13 Payment Instruction for Amount
S14 Notification of Deposit of Amount (In Case of Push Notification)
T40 Creation of Transaction B20
S15 Request for Registration of Transaction B20
6
7
8
9
10
11

Fig.6

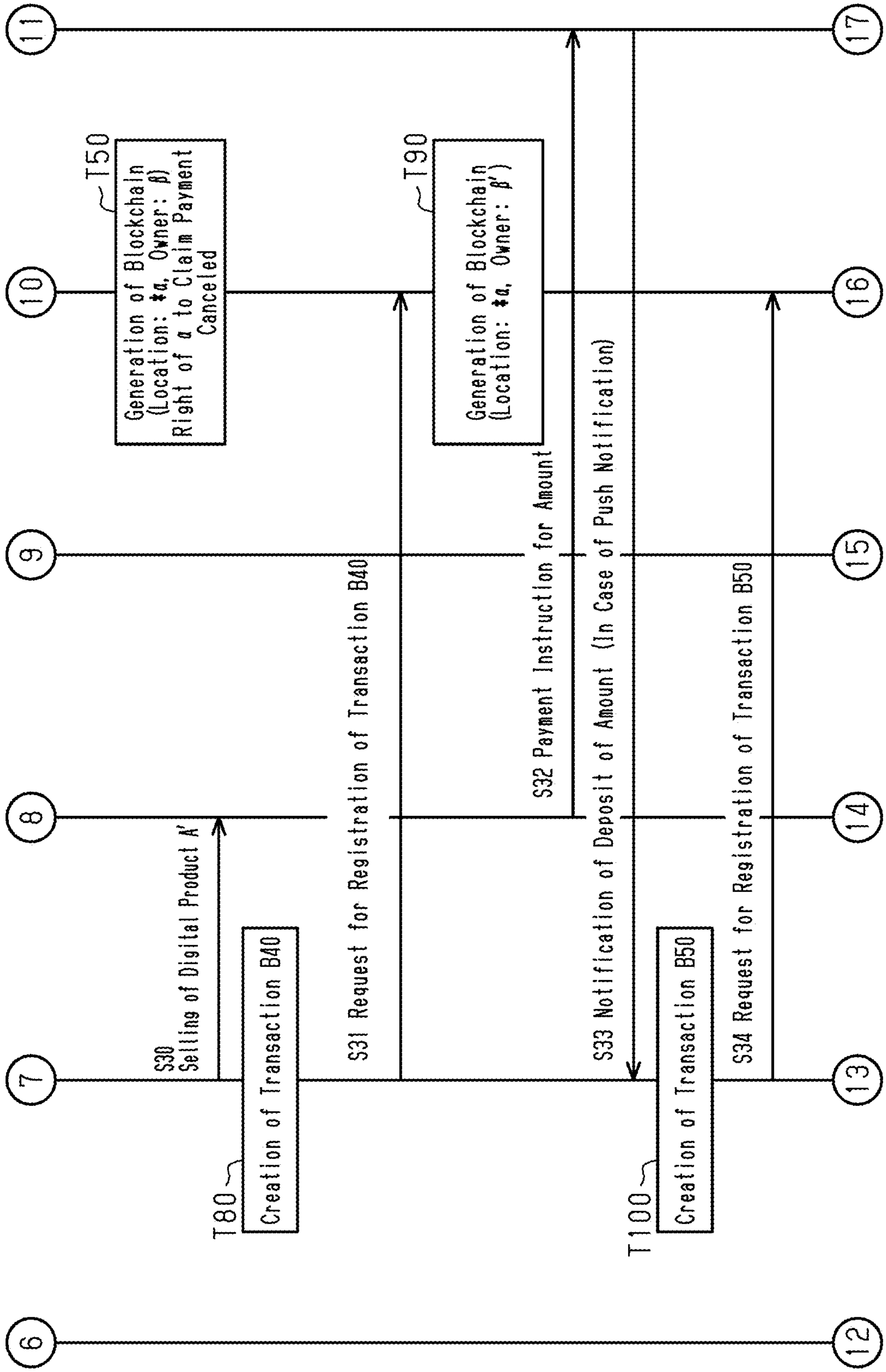
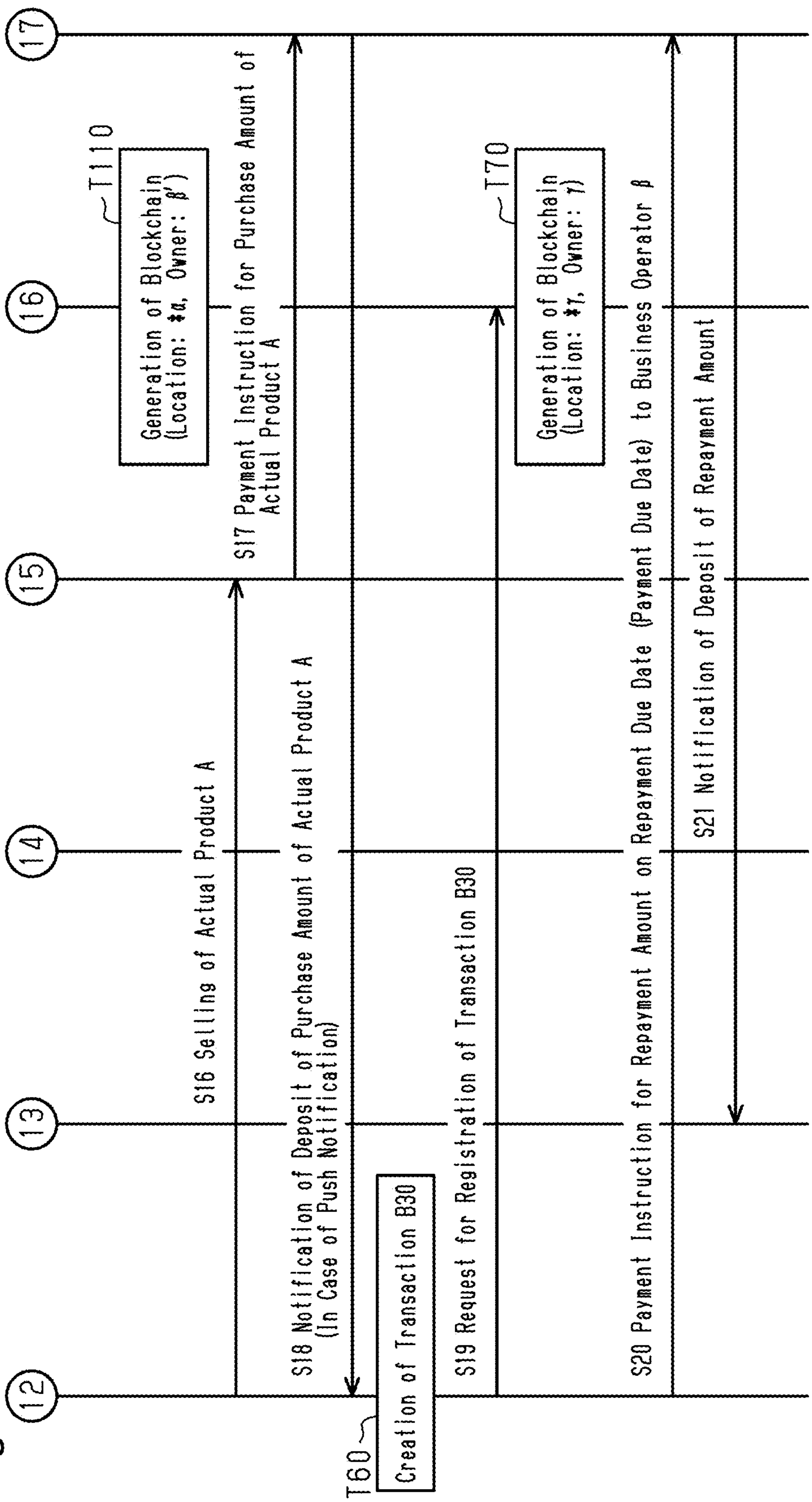

6
7
8
9
10
11
12
13
14
15
16
17
T50
Generation of Blockchain (Location: ‡α, Owner: β) Right of α to Claim Payment Canceled
T90
Generation of Blockchain (Location: ‡α, Owner: β')
S30 Selling of Digital Product A'
T80
Creation of Transaction B40
S31 Request for Registration of Transaction B40
S32 Payment Instruction for Amount
S33 Notification of Deposit of Amount (In Case of Push Notification)
T100
Creation of Transaction B50
S34 Request for Registration of Transaction B50

Fig.7


12
13
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17
T110
Generation of Blockchain (Location: #α, Owner: β')
S17 Payment Instruction for Purchase Amount of Actual Product A
T70
Generation of Blockchain (Location: #γ, Owner: γ)
S16 Selling of Actual Product A
S18 Notification of Deposit of Purchase Amount of Actual Product A (In Case of Push Notification)
T60
Creation of Transaction B30
S19 Request for Registration of Transaction B30
S20 Payment Instruction for Repayment Amount on Repayment Due Date (Payment Due Date) to Business Operator β
S21 Notification of Deposit of Repayment Amount

Fig.8

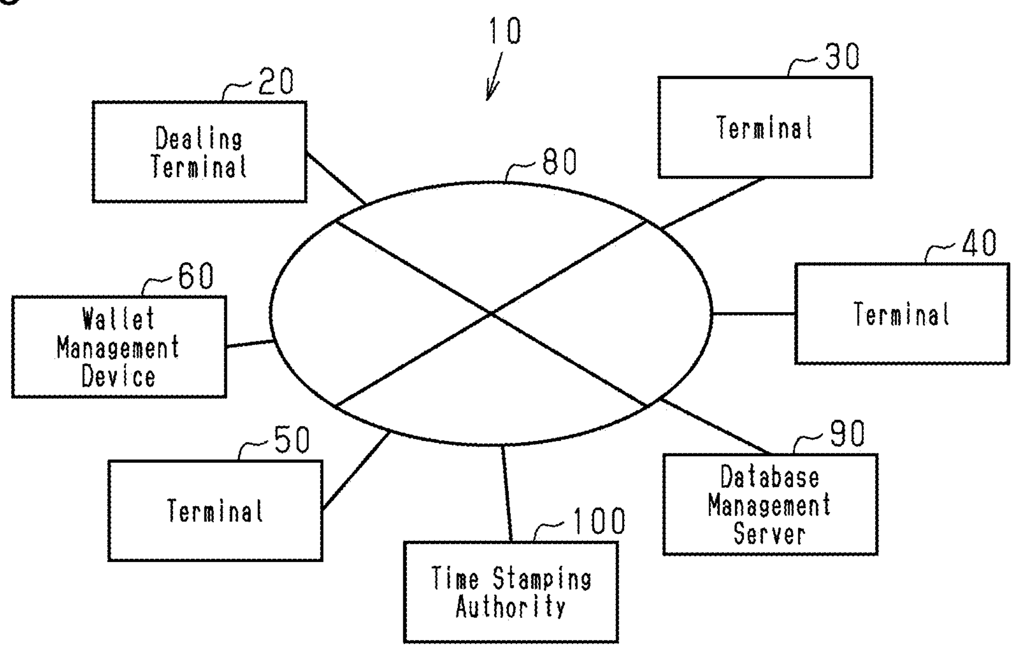
10
20
Dealing Terminal
30
Terminal
80
40
Terminal
60
Wallet Management Device
50
Terminal
90
Database Management Server
100
Time Stamping Authority

Fig.9

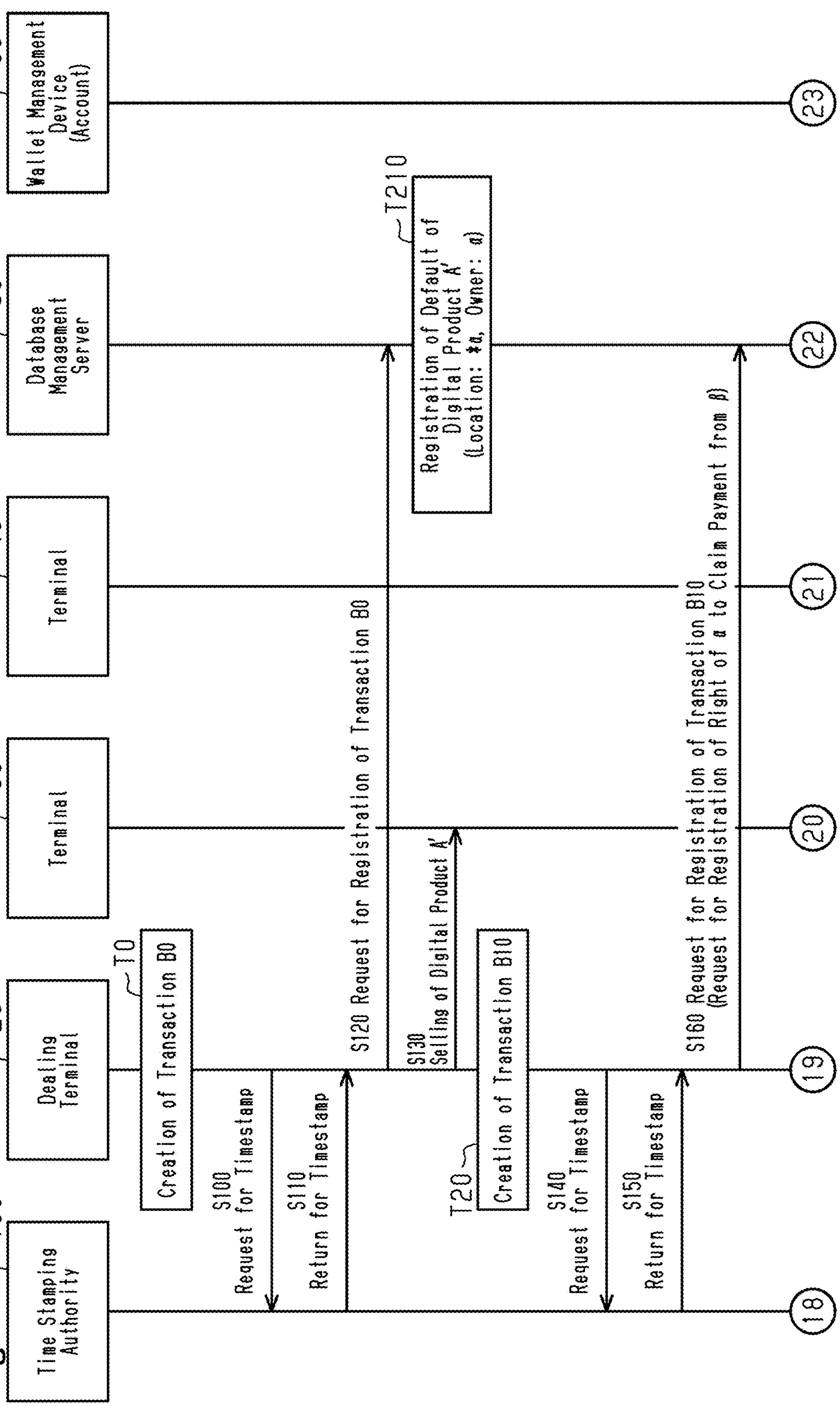
100
Time Stamping Authority
20
Dealing Terminal
30
Terminal
40
Terminal
90
Database Management Server
60
Wallet Management Device (Account)
T0
Creation of Transaction B0
S100 Request for Timestamp
S110 Return for Timestamp
S120 Request for Registration of Transaction B0
S130 Selling of Digital Product A'
T210
Registration of Default of Digital Product A' (Location: *α, Owner: α)
T20
Creation of Transaction B10
S140 Request for Timestamp
S150 Return for Timestamp
S160 Request for Registration of Transaction B10 (Request for Registration of Right of α to Claim Payment from β)
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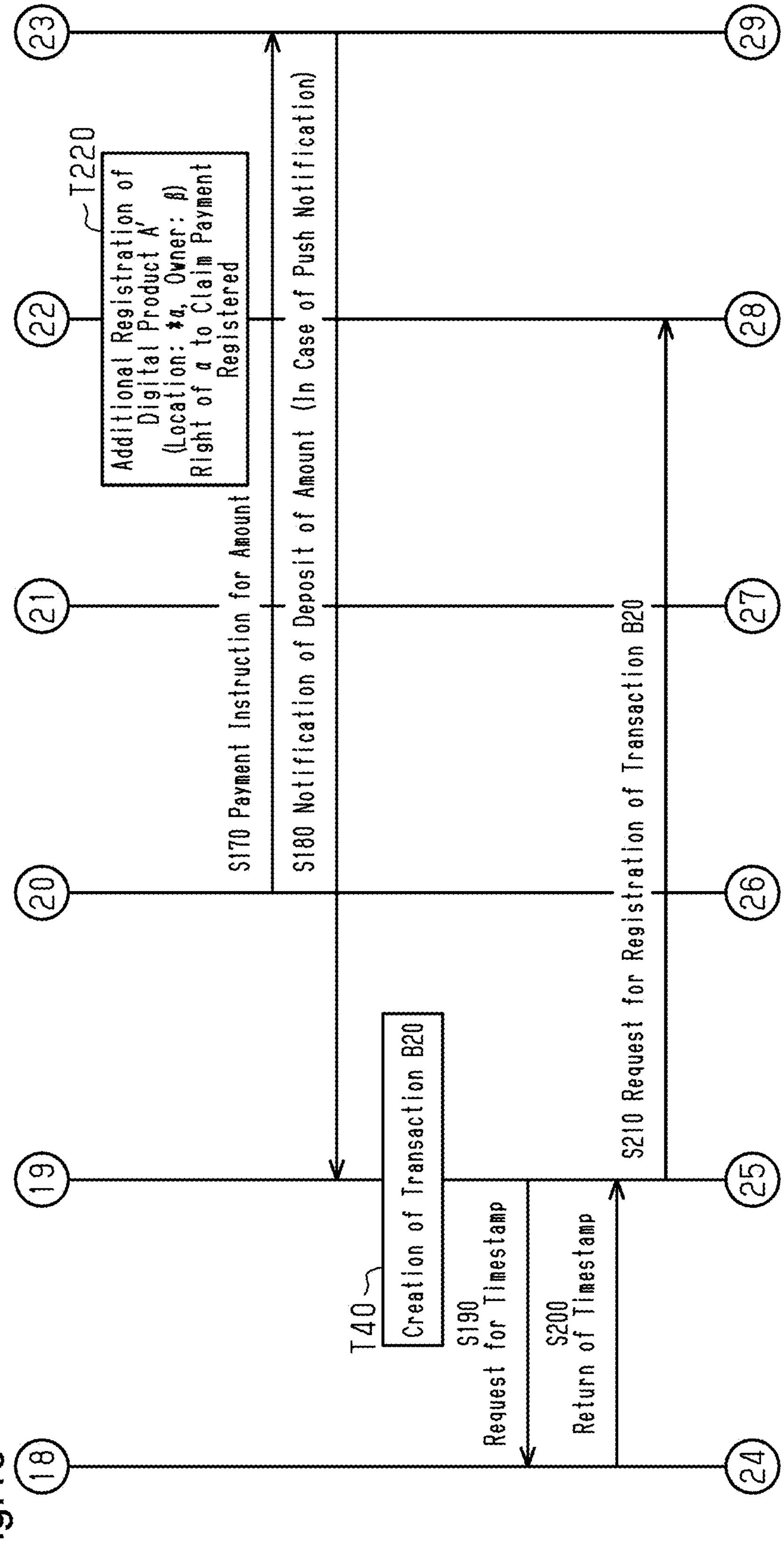
Fig.10


18
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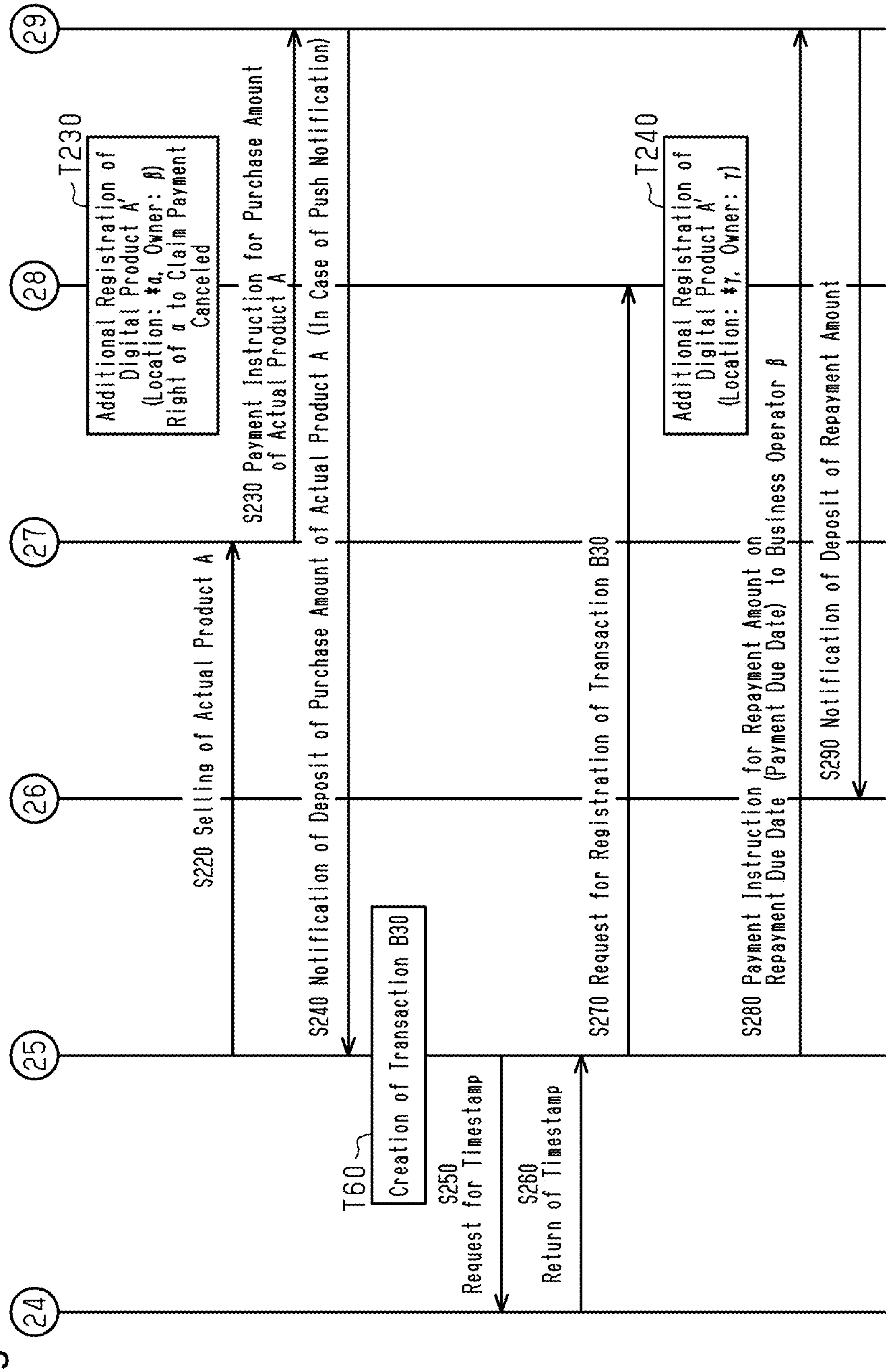
T220
Additional Registration of Digital Product A' (Location: #α, Owner: β) Right of α to Claim Payment Registered
S170 Payment Instruction for Amount
S180 Notification of Deposit of Amount (In Case of Push Notification)
T40
Creation of Transaction B20
S190 Request for Timestamp
S200 Return of Timestamp
S210 Request for Registration of Transaction B20

Fig.11


24
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T230
Additional Registration of Digital Product A' (Location: #α, Owner: β) Right of α to Claim Payment Canceled
S220 Selling of Actual Product A
S230 Payment Instruction for Purchase Amount of Actual Product A
S240 Notification of Deposit of Purchase Amount of Actual Product A (In Case of Push Notification)
T60
Creation of Transaction B30
S250 Request for Timestamp
S260 Return of Timestamp
S270 Request for Registration of Transaction B30
T240
Additional Registration of Digital Product A' (Location: #γ, Owner: γ)
S280 Payment Instruction for Repayment Amount on Repayment Due Date (Payment Due Date) to Business Operator β
S290 Notification of Deposit of Repayment Amount

Fig.12

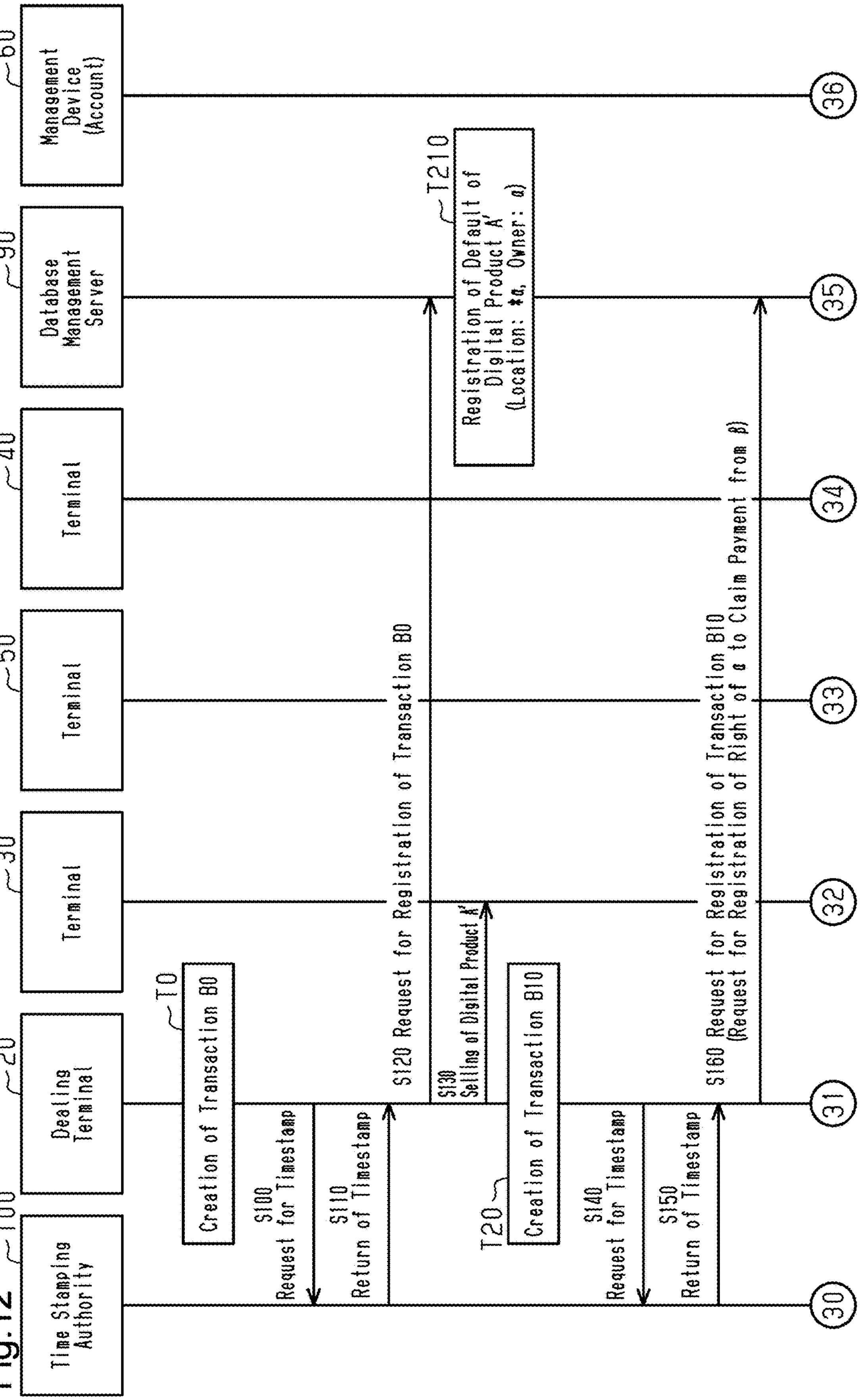
Time Stamping Authority
100
Dealing Terminal
20
Terminal
30
Terminal
50
Terminal
40
Database Management Server
90
Management Device (Account)
60
T0
Creation of Transaction B0
S100 Request for Timestamp
S110 Return of Timestamp
S120 Request for Registration of Transaction B0
S130 Selling of Digital Product A'
T210
Registration of Default of Digital Product A' (Location: *α, Owner: α)
T20
Creation of Transaction B10
S140 Request for Timestamp
S150 Return of Timestamp
S160 Request for Registration of Transaction B10 (Request for Registration of Right of α to Claim Payment from β)
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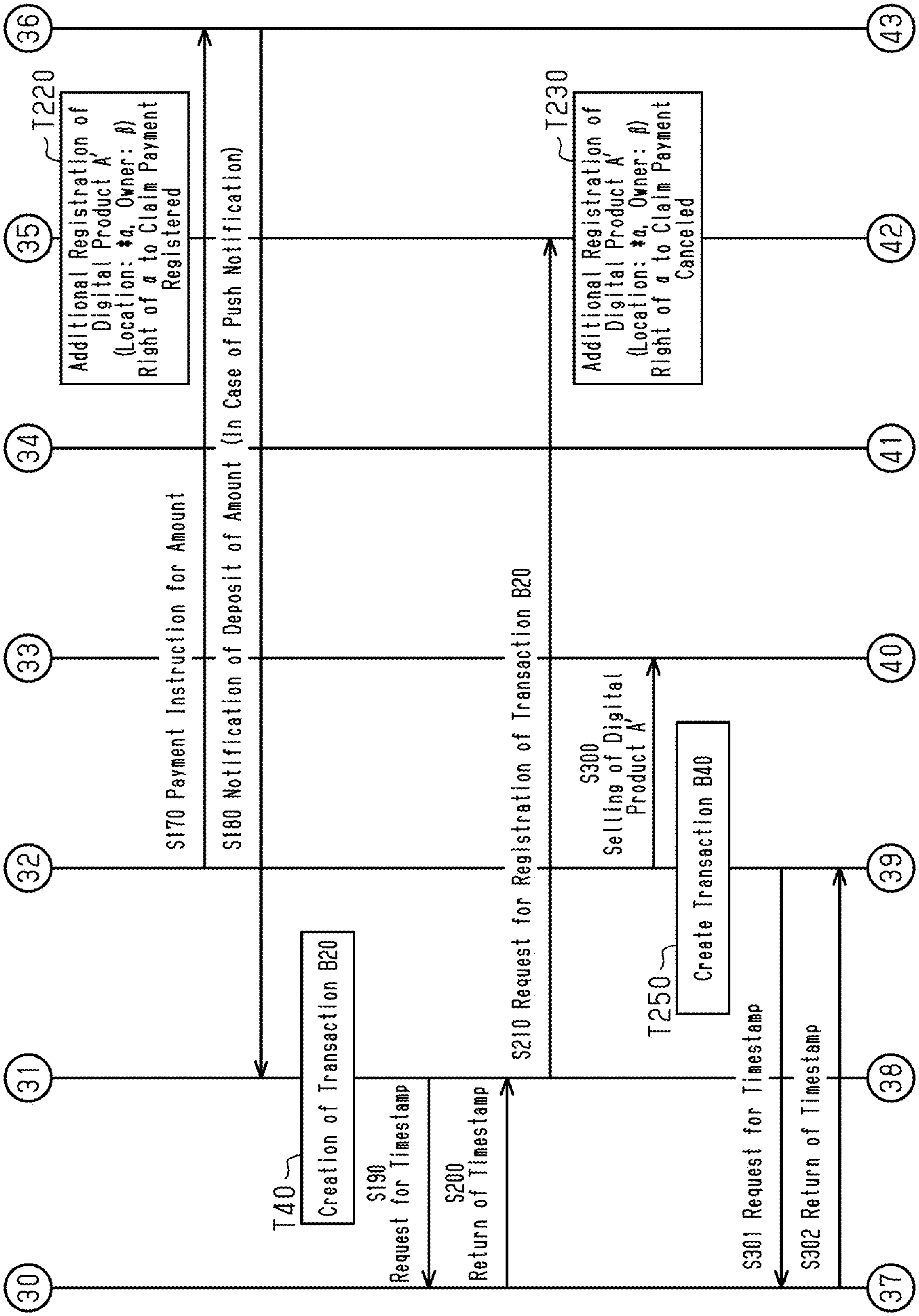
Fig.13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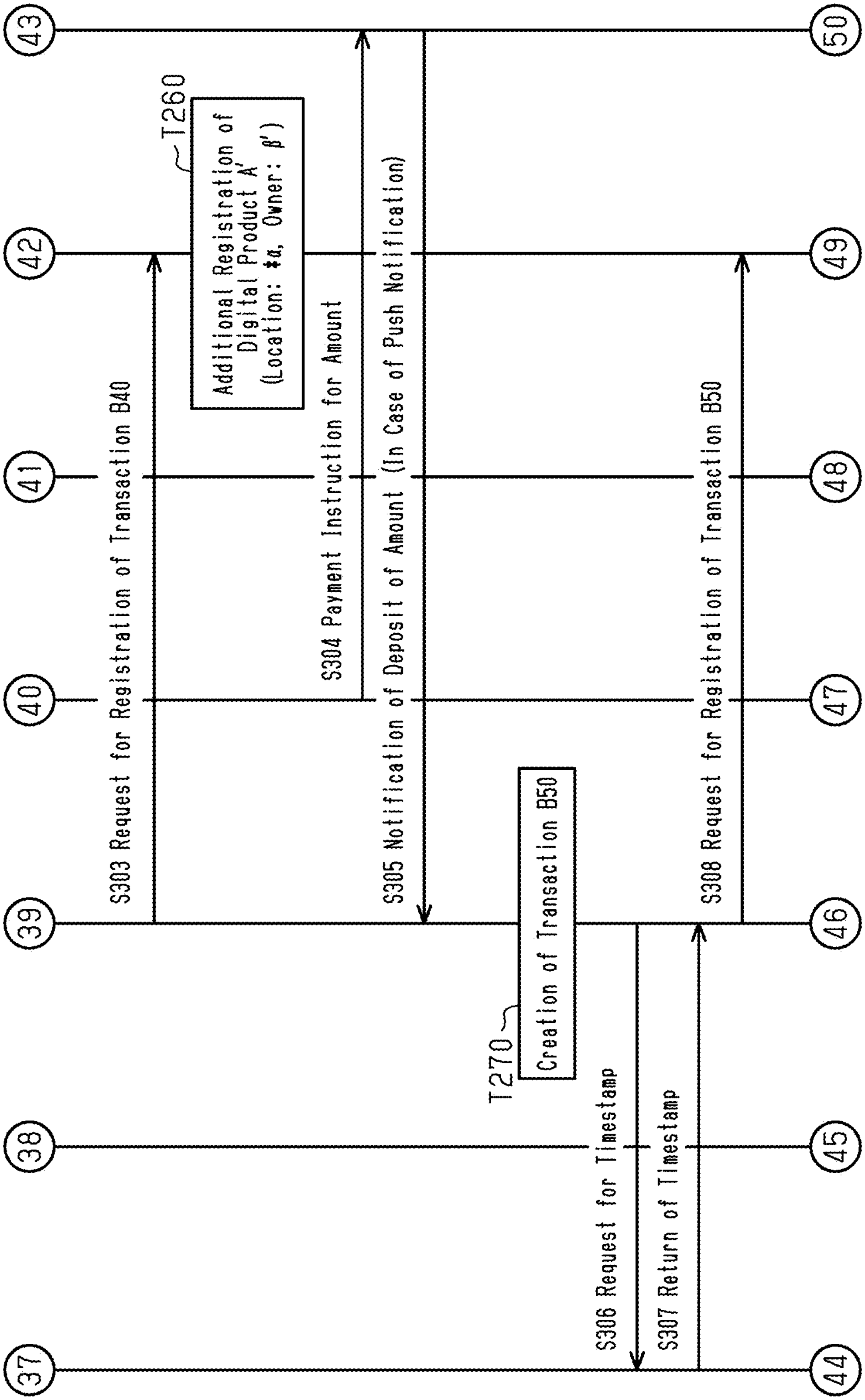
43
T220
Additional Registration of Digital Product A' (Location: #a, Owner: β) Right of a to Claim Payment Registered
S170 Payment Instruction for Amount
S180 Notification of Deposit of Amount (In Case of Push Notification)
T40
Creation of Transaction B20
S190 Request for Timestamp
S200 Return of Timestamp
S210 Request for Registration of Transaction B20
T230
Additional Registration of Digital Product A' (Location: #a, Owner: β) Right of a to Claim Payment Canceled
S300 Selling of Digital Product A'
T250
Create Transaction B40
S301 Request for Timestamp
S302 Return of Timestamp

Fig.14


37
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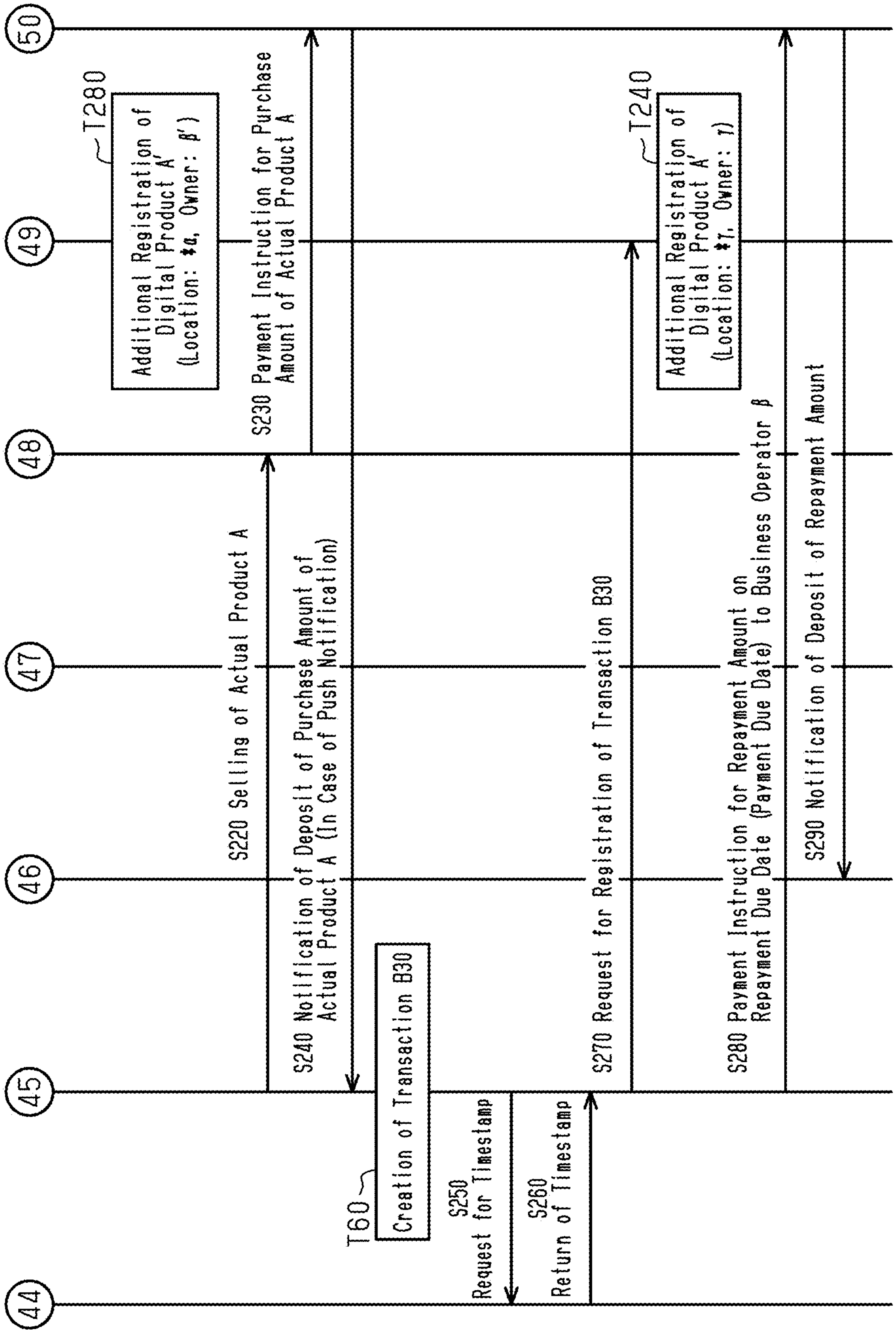
49
50
T260
Additional Registration of Digital Product A' (Location: #α, Owner: β')
S303 Request for Registration of Transaction B40
S304 Payment Instruction for Amount
S305 Notification of Deposit of Amount (In Case of Push Notification)
T270
Creation of Transaction B50
S306 Request for Timestamp
S307 Return of Timestamp
S308 Request for Registration of Transaction B50

Fig.15


44
45
46
47
48
49
50
T60
Creation of Transaction B30
S250 Request for Timestamp
S260 Return of Timestamp
S220 Selling of Actual Product A
S240 Notification of Deposit of Purchase Amount of Actual Product A (In Case of Push Notification)
S230 Payment Instruction for Purchase Amount of Actual Product A
T280
Additional Registration of Digital Product A' (Location: #α, Owner: β')
S270 Request for Registration of Transaction B30
T240
Additional Registration of Digital Product A' (Location: #γ, Owner: γ)
S280 Payment Instruction for Repayment Amount on Repayment Due Date (Payment Due Date) to Business Operator β
S290 Notification of Deposit of Repayment Amount

DIGITAL PRODUCT MANAGEMENT SUPPORT METHOD, DIGITAL PRODUCT MANAGEMENT SUPPORT DEVICE, DIGITAL PRODUCT MANAGEMENT SUPPORT PROGRAM, AND DIGITAL PRODUCT MANAGEMENT SUPPORT SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from prior Japanese Patent Application No. 2024-026345, filed on Feb. 26, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Field

[0002] The present disclosure relates to a digital product management support method, a digital product management support device, a digital product management support program, and a digital product management support system.

2. Description of Related Art

[0003] Fund raising methods include various methods such as bond issuance, borrowing, lending, and factoring. For example, in factoring disclosed in Japanese Laid-Open Patent Publication No. 2002-109224, a procedure for factoring accounts receivable is implemented by a computer digital product management support system.

[0004] In factoring, accounts receivable (receivables) generated when a business operator sells products are collateral. The business operator can raise funds when a factoring company purchases the accounts receivable. In this case, the business operator is provided with the amount obtained by subtracting a fee as funds from the factoring company.

[0005] After the funds are raised, the business operator is paid accounts receivable from purchasers of the products. Thereafter, the business operator pays these accounts receivable to the factoring company. In the e-commerce support system of the above document, it is possible to reduce the burden of clerical work and to promptly purchase accounts receivable.

[0006] In the above-described factoring, the issuance of an invoice after the sale of a product is a precondition. For this reason, funds cannot be raised until accounts receivable (receivables) are generated. A business operator having products may wish to raise funds before selling the products. However, with the factoring, it is difficult to raise funds before selling products.

[0007] An objective of the present disclosure is to provide a digital product management support method, a digital product management support device, a digital product management support program, and a digital product management support system that enable a business operator having a product to raise funds before the business operator sells the product.

SUMMARY

[0008] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0009] In accordance with a first aspect of the present disclosure, a digital product management support method for causing a computer of a dealing terminal to execute the following steps upon dealing of a digital product associated with an actual product is provided. The method includes: when there is a commercial dealing of the digital product, requesting, by the computer, a database entity under a communication network to register a transaction including owner information in which a purchaser of the digital product is a new owner of the digital product and a fact that a seller of the digital product has a right to claim payment for the digital product; in a case in which a deposit notification of a purchase amount of the digital product from the purchaser of the digital product is received via the communication network, requesting, by the computer, the database entity to register a transaction including a fact that the seller of the digital product does not have the right to claim payment; in a case in which the actual product is sold, requesting the database entity to register a transaction including a fact that the purchaser of the actual product is an owner of the digital product and a location of the actual product is registered as a location of the digital product; and on a payment due date according to a sales contract with the purchaser of the digital product based on the dealing of the digital product, notifying, by the computer, a terminal of a financial institution of payment of a sales amount of the digital product to an account of the purchaser of the digital product via the communication network.

[0010] In the present specification, dealing information of commercial dealing is referred to as a transaction.

[0011] In accordance with a second aspect of the present disclosure, a digital product management support device communicable with multiple terminals and a database entity via a communication network and configured to support management of dealing of a digital product associated with an actual product is provided. The device includes a first registration request unit, a second registration request unit, a third registration request unit, and a payment notification unit. The first registration request unit includes a processor configured to, when there is a commercial dealing of the digital product, request the database entity under the communication network to register a transaction including owner information in which a purchaser of the digital product is a new owner of the digital product and a fact that a seller of the digital product has a right to claim payment for the digital product. The second registration request unit includes a processor configured to, in a case in which a deposit notification of a purchase amount of the digital product from the purchaser of the digital product is received via the communication network, request the database entity to register a transaction including a fact that the seller of the digital product does not have the right to claim payment. The third registration request unit includes a processor configured to, in a case in which the actual product is sold, request the database entity to register a transaction including a fact that the purchaser of the actual product is an owner of the digital product and a location of the actual product is registered as a location of the digital product. The payment notification unit includes a processor configured to request a terminal of a financial institution to pay a sales amount of the digital product to an account of the purchaser of the digital product via the communication network on a payment due date according to a sales contract with the purchaser of the digital product based on the dealing of the digital product.

[0012] In accordance with a third aspect of the present disclosure, a computer-readable medium recording a digital product management support program implemented by multiple terminals and a computer communicable with a database entity via a communication network is provided. The digital product management support program includes an instruction that causes the computer to: when there is a commercial dealing of a digital product associated with an actual product, request the database entity under the communication network to register a transaction including owner information in which a purchaser of the digital product is a new owner of the digital product and a fact that a seller of the digital product has a right to claim payment for the digital product; in a case in which a deposit notification of a purchase amount of the digital product from the purchaser of the digital product is received via the communication network, request the database entity to register a transaction including a fact that the seller of the digital product does not have the right to claim payment; in a case in which the actual product is sold, request the database entity to register a transaction including a fact that the purchaser of the actual product is an owner of the digital product and a location of the actual product is registered as a location of the digital product; and notify a terminal of a financial institution of payment of a sales amount of the digital product to an account of the purchaser of the digital product via the communication network on a payment due date according to a sales contract with the purchaser of the digital product based on the dealing of the digital product.

[0013] In accordance with a fourth aspect of the present disclosure, a digital product management support system is provided that includes a digital product management support device, multiple terminals, and a database entity that are connected via a communication network. The digital product management support device includes a first registration request unit, a second registration request unit, a third registration request unit, and a payment notification unit. The first registration request unit includes a processor configured to, when there is a commercial dealing of a digital product associated with an actual product, request the database entity under the communication network to register a transaction including owner information in which a purchaser of the digital product is a new owner of the digital product and a fact that a seller of the digital product has a right to claim payment for the digital product. The second registration request unit includes a processor configured to, in a case in which a deposit notification of a purchase amount of the digital product from the purchaser of the digital product is received via the communication network, request the database entity to register a transaction including a fact that the seller of the digital product does not have the right to claim payment. The third registration request unit includes a processor configured to, in a case in which the actual product is sold, request the database entity to register a transaction including a fact that the purchaser of the actual product is an owner of the digital product and a location of the actual product is registered as a location of the digital product. The payment notification unit includes a processor configured to request a terminal of a financial institution to pay a sales amount of the digital product to an account of the purchaser of the digital product via the communication network on a payment due date according to a sales contract with the purchaser of the digital product based on the dealing of the digital product. The database entity is configured to perform processing according to a registration request of the first registration request unit, a registration request of the second registration request unit, and a registration request of the third registration request unit.

[0014] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. **1** is a block diagram illustrating a configuration of a digital product management support system according to a first embodiment of the present disclosure.

[0016] FIG. **2** is a block diagram illustrating a configuration of each of a dealing terminal and a terminal of the first embodiment.

[0017] FIG. **3** is a sequence diagram illustrating an outline of digital product management support of the first embodiment.

[0018] FIG. **4** is a sequence diagram illustrating the outline of the digital product management support of the first embodiment.

[0019] FIG. **5** is a sequence diagram illustrating an outline of digital product management support of a second embodiment.

[0020] FIG. **6** is a sequence diagram illustrating the outline of the digital product management support of the second embodiment.

[0021] FIG. **7** is a sequence diagram illustrating the outline of the digital product management support of the second embodiment.

[0022] FIG. **8** is a block diagram illustrating a configuration of a digital product management support system according to a third embodiment.

[0023] FIG. **9** is a sequence diagram illustrating an outline of the digital product management support of the third embodiment.

[0024] FIG. **10** is a sequence diagram illustrating the outline of the digital product management support of the third embodiment.

[0025] FIG. **11** is a sequence diagram illustrating the outline of the digital product management support of the third embodiment.

[0026] FIG. **12** is a sequence diagram illustrating an outline of digital product management support of a fourth embodiment.

[0027] FIG. **13** is a sequence diagram illustrating the outline of the digital product management support of the fourth embodiment.

[0028] FIG. **14** is a sequence diagram illustrating the outline of the digital product management support of the fourth embodiment.

[0029] FIG. **15** is a sequence diagram illustrating the outline of the digital product management support of the fourth embodiment.

[0030] Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

[0031] This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.

[0032] Exemplary embodiments may have different forms, and are not limited to the examples described. However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.

[0033] In this specification, "at least one of A and B" should be understood to mean "only A, only B, or both A and B."

First Embodiment

[0034] Hereinafter, a first embodiment embodying the present disclosure will be described with reference to FIGS. **1** to **4**.

1. Digital Product Management Support System

[0035] As illustrated in FIG. **1**, a digital product management support system **10** includes a dealing terminal **20**, terminals **30**, **40**, and **50**, a wallet management device **60**, and a distributed ledger system **70**. Although three terminals **30**, **40**, and **50** are shown for illustrative purposes, four or more terminals may be provided.

[0036] The dealing terminal **20**, the terminals **30**, **40**, and **50**, the wallet management device **60**, and the distributed ledger system **70** are connected via a communication network **80**. The communication network **80** includes, for example, the Internet, a wide area network (WAN), and the like.

1.1. Dealing Terminal 20, Terminals 30, 40, 50

[0037] In the present embodiment, the dealing terminal **20** and the terminals **30**, **40**, and **50** are formed by personal computers. The dealing terminal **20** and the terminals **30**, **40**, and **50** may be formed by smartphones or tablet terminals. As illustrated in FIG. **2**, the dealing terminal **20** includes a processor **22**, a storage unit **24**, an input/output interface **26**, and a communication interface **28**. The dealing terminal **20** corresponds to a digital product management support device. A personal computer forming the dealing terminal **20** corresponds to a computer of the present disclosure.

[0038] The processor **22** includes memories such as a central processing unit (CPU), a read-only memory (ROM), and a random access memory (RAM) (not illustrated). The processor **22** includes a ROM (not illustrated) that stores an operating system (OS) program for managing the entire system of the device that is the dealing terminal **20** itself, a RAM (not illustrated) serving as a work memory, and a CPU (not illustrated).

[0039] Instead of the CPU, an application specific integrated circuit (ASIC) or a field programmable array (FPGA) may be used. The processor **22** corresponds to a first registration request unit, a second registration request unit, a third registration request unit, and a payment notification unit.

[0040] The storage unit **24** is formed by a rewritable storage medium such as a hard disk or a solid state drive (SSD). The storage unit **24** stores a digital product management support program, an application program such as a web browser, and various data acquired by execution processing of the digital product management support program.

[0041] Various input devices and output devices (both not illustrated) are connected to the input/output interface **26**. The input devices include a keyboard, a touch panel, and a pointing device such as a mouse. The output devices include a display, a speaker, and a printer.

[0042] The communication interface **28** controls communication between the dealing terminal **20** and the wallet management device **60** and the distributed ledger system **70** via the communication network **80**. The communication interface **28** corresponds to a communication unit.

[0043] Since the terminals **30**, **40**, and **50** have the same configuration as that of the dealing terminal **20**, as illustrated in FIG. **2**, "(30, 40, 50)" is added after the reference numeral "20" of the dealing terminal **20**, and the description thereof is omitted.

[0044] Business operators who use the dealing terminal **20** and the terminals **30**, **40**, and **50** are both users who are qualified to participate in the distributed ledger system **70**. The business operators are individuals or corporations. These business operators can check the content of transactions of blocks of a blockchain by inputting the respective identifiers and passwords of the business operators using the input devices of the dealing terminal **20** and the terminals **30**, **40**, and **50**.

[0045] The password is preferably a public key cryptosystem.

1.2. Wallet Management Device 60

[0046] The wallet management device **60** is a management server of a financial institution such as a bank. In FIG. **2**, for illustrative purposes, one wallet management device **60** is representatively shown, but it should be understood that multiple wallet management devices **60** are connected to the communication network **80**. A case will now be described in which a payment instruction from an account of a business operator to an account of another business operator has been issued from any one of the dealing terminal **20** and the terminals **30**, **40**, and **50** to one wallet management device **60** due to a commercial dealing. Then, the wallet management device **60** remits money from the account of the business operator who has given the payment instruction to another wallet management device that manages the account of the other business operator according to the payment instruction.

[0047] In this manner, in a commercial dealing of a product and a digital product associated with the product, which will be described later, the wallet management device **60** manages an account of a purchaser used for payment for the amounts of the product and the digital product.

1.3. Distributed Ledger System 70

[0048] In the distributed ledger system **70**, multiple node computers are connected to each other wirelessly or by wire via a P2P network. The P2P network is a single configuration such as a public network, a dedicated line, or a virtual private network (VPN), or a combination thereof. The node computers each include a processor, a storage unit, an input/output interface, and a communication interface similar to those of the dealing terminal **20** illustrated in FIG. **2**. In the present embodiment, the dealing terminal **20** and the terminals **30**, **40**, and **50** are also included in the node computers forming the distributed ledger system **70**.

[0049] The distributed ledger system **70** corresponds to a database entity. Each of the node computers manages a distributed ledger in a blockchain. In other words, when acquiring the transaction for which registration has been requested, the node computer generates a block including the transaction, and links the generated block to an already-generated blockchain in a chain manner. As a result, the blockchain has a data structure in which the generated blocks are connected in time series. Examples of the blockchain include bitcoin (registered trademark) and Ethereum (registered trademark).

[0050] Hereinafter, a consensus algorithm will be described using proof of work (PoW) as an example. The consensus algorithm may be proof of stake (POS) or proof of importance (PoI).

[0051] A block has an area for storing various data such as an identifier of the block, a transaction acquired at a generation time interval of the block, a timestamp at which the transaction is acquired in the block, and each hash value of the block and the block generated last time.

[0052] The identifier is a serial number that is added one by one each time a new block is connected to the blockchain. By referring to the latest hash value of the blockchain, it is possible to publicly verify whether the blockchain has been tampered.

[0053] The block has multiple transactions. The block is time-series data in which multiple transactions are arranged according to the timestamp of the transaction. In this manner, the transaction is associated with a timestamp.

[0054] An actual product and a digital product can be sold separately. The actual product and the digital product are associated with each other. When the actual product is sold and the deposit of the sales price of the actual product is confirmed, the ownership of the digital product is transferred to the purchaser of the actual product.

[0055] The node computer generates a block by successfully searching (mining) for a nonce satisfying a predetermined condition every predetermined time. Blocks generated by one node computer are chained into an existing blockchain upon approval by another node computer. The blockchain thus updated is shared by the node computers connected to the P2P network. As a result, the blockchain becomes a database that stores multiple transactions (dealing information (including a dealing history)) in the P2P network.

[0056] A mining reward is given to a node computer that has first generated a block by successfully searching for a nonce. Examples of the mining reward include, but are not limited to, a cryptographic asset. For example, a mining reward may be paid by a business operator who manages a block chain.

Database by Distributed Ledger System **70**

[0057] Registration of a transaction in the distributed ledger system **70** forms a database related to an actual product and a digital product to be sold.

[0058] The database includes, for each transaction, transaction related information and digital product related information to be described later. The transaction related information and the digital product related information for each transaction will be described later.

Operation of First Embodiment

[0059] Next, operation of the digital product management support system **10**, the dealing terminal **20**, and the digital product management support program of the present embodiment will be described with reference to FIGS. **3** and **4**. FIGS. **3** and **4** are sequence diagrams illustrating an outline of digital product management support.

[0060] Hereinafter, an actual product is denoted by a reference sign A. A digital product is denoted by a reference sign A". In addition, *α is attached to the location of an actual product. A digital product A' is collateralized by its actual product A.

[0061] In addition, a is assigned to the business operator of the dealing terminal **20**, β is assigned to the business operator of the terminal **30**, γ is assigned to the business operator of the terminal **40**, and β" is assigned to the business operator of the terminal **50**. In addition, the business operator who uses the dealing terminal **20** is referred to as a vendor who sells the actual product A and the digital product A". The vendor is a business operator who raises funds by selling the digital product A". The business operator who uses the terminal **30** is a business operator who purchases the digital product A" to fund the vendor. The business operator γ who uses the terminal **40** is a purchaser of the actual product A.

T**0**: Creation of Transaction B**0**

[0062] As illustrated in FIG. **3**, the dealing terminal **20** of the vendor (business operator α) creates a transaction B**0**. The creation of a transaction B**10** is performed by an input operation or the like of an input device of the dealing terminal **20**. The transaction B**0** includes the following (1) transaction related information, (2) digital product related information, and (3) actual product related information.

[0063] The business operator α assigns individual management (unique serial number) for managing a product such as an RFID tag or a QR code (registered trademark) to the actual product A, and manages the stock of the actual product A and the digital product A'. That is, the actual product A and the digital product A' are managed in a one-to-one linked state to avoid overlapping.

Transaction B**0**

[0064] (1) Since the sale has not been completed, the various types of information included in the transaction related information is default data unlike the transaction B**10** to be described later.

[0065] (2) Digital product related information: an image and an identifier of the digital product A', the owner of the digital product A', the purchase amount (trade price) of the digital product A', and the location of the digital product A'. The image of the digital product A' only needs to be able to identify the digital product A', and is not necessarily limited to a captured image of the actual product A.

[0066] (3) Actual product related information: the name of the actual product A associated with the digital product A', the identifier of the actual product A, the cost price of the actual product A, the sales price of the actual product A, and the location of the actual product A.

[0067] In B**0**, the owner of the digital product A' is the vendor (business operator α). The location *α of the digital product A' is the same as the location of the actual product A. Therefore, in B**0**, the location of the digital product A' is the location of the vendor who owns the actual product A. The location of the digital product A' corresponds to the location information of the digital product A'. The owner of the digital product A' corresponds to the owner information of the digital product A'.

S**10**: Request Registration of Transaction B**0**

[0068] As described above, when the transaction B**0** is created, the digital product management support program causes the dealing terminal **20** to request the distributed ledger system **70** to register the transaction B**0** via the communication network **80**.

T**10**: Generation of Blockchain

[0069] In the distributed ledger system **70**, when the registration of the transaction B**0** is requested, generation of a block including the transaction B**0** and generation of a blockchain are performed. As a result, the location *α of the digital product A' and the fact that the owner of the digital product A' is the vendor (business operator α) are registered in the database of the distributed ledger system **70**.

S**11**: Selling of Digital Product A'

[0070] As illustrated in FIG. **3**, when a sales contract is established between the vendor (business operator α) and a business operator (business operator β) who wishes to purchase the digital product A', digital data including an image of the digital product A' and the like, and a delivery slip and an invoice are transmitted from the dealing terminal **20** to the terminal **30**. This transmission is executed by the processor **22** of the dealing terminal **20** when an input device connected to the input/output interface **26** is operated. The delivery of the digital data including the image of the digital product A' and the like, and the delivery of the delivery slip and the invoice from the dealing terminal **20** to the terminal **30** means the sale of the digital product A'.

[0071] The commercial dealing between the business operator α and the business operator β may be either face-to-face dealing or dealing on a website provided by a server (not illustrated) connected to the communication network **80**.

[0072] Therefore, the business operator β can confirm the content of the transaction of the block of the blockchain by inputting the identifier and the password of the business operator β using the input devices of the terminal **30**. That is, the business operator β can confirm the content of the digital product A' when purchasing the digital product A'.

T**20**: Creation of Transaction B**10**

[0073] As illustrated in FIG. **3**, after the sale of the digital product A' is completed, the dealing terminal **20** creates the transaction B**10**. The creation of the transaction B**10** is performed by an input operation or the like of the input device of the dealing terminal **20**.

[0074] The transaction B**10** includes the following (1) transaction related information, (2) digital product related information, and (3) actual product related information.

Transaction B**10**

[0075] (1) Transaction related information: an identifier of a transaction, and sales contract content information between the business operators α and β.

[0076] The contract information includes the following information.

[0077] Identifiers of the business operators α and β, dealing conclusion date and time (purchase date and time), a payment due date of a purchase amount of the digital product A' from the business operator β to the business operator α, a payment due date of a repayment amount X from the business operator α to the business operator β, a payment rate π to a purchaser (business operator β) of the digital product A', information indicating that a seller of the digital product A' has a right to claim payment (monetary claim) for the digital product A', and the like.

[0078] (2) The digital product related information includes an image and an identifier of the digital product A', owner information of the digital product A', the location *α of the digital product A', and the purchase amount of the digital product A' (trade price D: $D=X(1-\pi)$).

[0079] X is the price of the digital product A' and the repayment amount from the vendor (business operator α) to the purchaser (business operator β) of the digital product A'. π is the payment rate, that is, the discount rate from the vendor (business operator α) to the purchaser of the digital product A'.

[0080] In the transaction B**10**, it is indicated that the owner of the digital product A' is the business operator β. The owner of the digital product A' corresponds to the owner information of the digital product A'.

[0081] (3) Actual product related information: the name of the actual product A associated with the digital product, the identifier of the actual product A, the cost price of the actual product A, the sales price of the actual product A, and the location of the actual product A.

[0082] In the transaction B**10**, the location of the actual product A is the location *α of the vendor who owns the actual product A.

S**12**: Request for Registration of Transaction B**10**

[0083] As described above, when the transaction B**10** is created, the digital product management support program causes the dealing terminal **20** to request the distributed ledger system **70** to register the transaction B**10** via the communication network **80**.

[0084] The registration request includes a registration request of the owner information of the digital product A' in which the purchaser of the digital product A' is the new owner of the digital product A', and a registration request indicating that the seller of the digital product A' has the right to claim payment for the digital product A'.

T**30**: Generation of Blockchain

[0085] In the distributed ledger system **70**, when registration of the transaction B**10** is requested, generation of a block including the transaction B**10** and generation of a blockchain are performed.

[0086] By the transaction B**10**, the location *α of the digital product A', information indicating that the owner of the digital product A' is the purchaser (business operator β) of the digital product A', and that the seller of the digital product A' has the right to claim payment for the digital product A' are registered in the database of the distributed ledger system **70**.

[0087] The fact that the transaction is associated with the timestamp means that the location *α and the owner information of the digital product A' are associated with the timestamp. It should be noted that, in the following description of the generation of the block chain, the location and the owner information of the digital product A' are associated with the timestamp added to the transaction in this manner.

S**13**: Payment Instruction for Amount

[0088] In S**13**, the terminal **30** of the business operator β instructs the wallet management device **60** to pay the purchase amount (trade price D) of the digital product A' to the vendor (business operator α) via the communication network **80**. In response to the payment instruction, the wallet management device **60** remits money from the account of the business operator who has given the payment instruction to another wallet management device that manages the account of the vendor (business operator α).

S**14**: Notification of Deposit of Amount

[0089] In S**14**, a deposit notification (for example, push notification) of the amount from the business operator β is received via the other wallet management device to the account of the vendor via the communication network **80**. The currency to be deposited in the account is a stablecoin or cash. A stablecoin is a currency for which price fluctuation is stabilized by having legal currency, virtual currency, gold, or the like as collateral. Wallet depositing in the following other steps is similarly performed with a stablecoin or cash.

T**40**: Creation of Transaction B**20**

[0090] As illustrated in FIG. **3**, when a notification of deposit of the amount from the business operator β to the account of the vendor is received via the communication network **80**, the digital product management support program causes the dealing terminal **20** to create a transaction B**20**.

Transaction B**20**

[0091] The transaction B**10** has information indicating that the seller of the digital product A' has the right to claim payment for the digital product A'. On the other hand, the transaction B**20** has information indicating that the seller of the digital product A' does not have the right to claim payment for the digital product A'. In this respect, the transaction B**20** is different from the transaction B**10**. In the transaction B**20**, other information except for the information indicating that there is no right to claim payment for the digital product A', that is, the transaction related information, the digital product related information, and the actual product related information are similar to those in the transaction B**10**, and thus, description thereof is omitted.

[0092] Therefore, in the transaction B**20**, similarly to the transaction B**10**, the owner of the digital product A' is the purchaser (business operator β), and there is no change from the transaction B**10**. In the transaction B**20**, the location *α of the digital product A' is not changed from the transaction B**10**.

[0093] The location *α of the digital product A' corresponds to the location information of the digital product A'.

S**15**: Request for Registration of Transaction B**20**

[0094] As described above, when the transaction B**20** is created, the digital product management support program causes the dealing terminal **20** to request the distributed ledger system **70** to register the transaction B**20** via the communication network **80**.

[0095] The registration request includes a registration request indicating that the seller of the digital product A' does not have the right to claim payment for the digital product A'.

T**50**: Generation of Blockchain

[0096] In T**50**, in the distributed ledger system **70**, when registration of the transaction B**20** is requested, generation of a block including the transaction B**20** and generation of a blockchain are performed.

[0097] By the transaction B**20**, the location *α of the digital product A', information indicating that the owner of the digital product A' is the purchaser (business operator β) of the digital product A', and that the seller of the digital product A' does not have the right to claim payment for the digital product A' are registered in the database of the distributed ledger system **70**.

S**16**: Selling of Actual Product A

[0098] As illustrated in FIGS. **3** and **4**, in a case in which a commercial dealing between the vendor (business operator α) and a business operator (business operator γ) who wishes to purchase the actual product A, that is, a sales contract is established, a delivery slip and an invoice of the actual product A are transmitted from the dealing terminal **20** to the terminal **40**. This transmission is executed by the processor **22** of the dealing terminal **20** when an input device connected to the input/output interface **26** is operated. The delivery of the delivery slip and the invoice of the actual product A from the dealing terminal **20** to the terminal **40** means the sale of the actual product A.

[0099] The commercial dealing between the business operator α and the business operator γ may be either face-to-face dealing or dealing on a website provided by a server (not illustrated) connected to the communication network **80**.

S**17**: Payment Instruction for Purchase Amount of Actual Product A

[0100] In S**17**, the terminal **40** of the business operator γ instructs the wallet management device **60** to pay the purchase amount (trade price D) of the actual product A to the vendor (business operator α) via the communication network **80**. In response to the payment instruction, the wallet management device **60** remits money from the account of the business operator who has given the payment instruction to another wallet management device that manages the account of the vendor (business operator α).

S**18**: Notification of Deposit of Purchase Amount of Actual Product A

[0101] In S**18**, a notification (for example, push notification) of deposit of the amount from the business operator γ is received via the other wallet management device to the account of the vendor via the communication network **80**.

T**60**: Creation of Transaction B**30**

[0102] When a notification of deposit of the amount from the business operator γ to the account of the vendor is received via the communication network **80**, the digital product management support program causes the dealing terminal **20** to create a transaction B**30**.

[0103] The transaction B**30** includes the following (1) transaction related information, (2) digital product related information, and (3) actual product related information.

Transaction B**30**

[0104] (1) Transaction related information: an identifier of a transaction, and sales contract content information between the business operators α and β (including identifiers of a vendor and a purchaser, a dealing conclusion date and time (purchase date and time), a payment date of a purchase amount of the actual product A from the purchaser to the vendor, and the like). In addition, the transaction related information also includes information indicating that there is no right to claim payment for the digital product A'.

[0105] (2) Digital product related information: an image and an identifier of the digital product A', the owner of the digital product A', the purchase amount (trade price) of the digital product A', and the location of the digital product A'. In S**18**, the notification of the deposit of the amount from the business operator γ is confirmed. As a result, the ownership of the digital product A' is transferred to the purchaser (business operator γ) of the actual product A. Therefore, in the transaction B**30**, the owner of the digital product A' is the purchaser of the actual product A (business operator γ), and the location of the digital product A' is the location of the purchaser (business operator γ) who has purchased the actual product A from the vendor.

[0106] (3) Actual product related information: the name of the actual product A associated with the digital product, the identifier of the actual product A, the cost price of the actual product A, the sales price of the actual product A, and the location of the actual product A.

S**19**: Request for Registration of Transaction B**30**

[0107] As described above, when the transaction B**30** is created, the digital product management support program causes the dealing terminal **20** to request the distributed ledger system **70** to register the transaction B**30** via the communication network **80**.

[0108] The registration request includes a registration request of owner information of the digital product A' in which the purchaser (business operator γ) of the actual product A is a new owner of the digital product A', and a registration request of a location of the digital product A'.

T**70**: Generation of Blockchain

[0109] In the distributed ledger system **70**, when registration of the transaction B**30** is requested, generation of a block including the transaction B**30** and generation of a blockchain are performed.

[0110] By the transaction B**30**, the location *γ of the digital product A' and the fact that the owner of the digital product A' is the purchaser (business operator γ) of the actual product A are registered in the database of the distributed ledger system **70**.

S**20**: Payment Instruction for Repayment Amount on Repayment Due Date (Payment Due Date) to Business Operator β

[0111] In S**20**, the following is executed by the digital product management support program. That is, on the payment due date of the repayment amount X from the dealing terminal **20** of the business operator α to the purchaser (business operator β) of the digital product A', the wallet management device **60** is instructed to pay the repayment amount of the digital product A' to the business operator β via the communication network **80**.

[0112] In response to the payment instruction, the wallet management device **60** remits money from the account of the business operator who has given the payment instruction to another wallet management device that manages the account of the business operator β.

S**21**: Notification of Deposit of Repayment Amount

[0113] In S**21**, a deposit notification (for example, push notification) of the repayment amount from the business operator β is received via the other wallet management device to the terminal **30** of the business operator β via the communication network **80**.

[0114] The first embodiment has the following features.

[0115] (1) A digital product management support method of the present embodiment has the following features.

[0116] In S**12**, the dealing terminal **20** requests the distributed ledger system **70** to register a transaction including owner information of the digital product A' whose new owner is the purchaser (business operator β) of the digital product A' and the fact that a seller of the digital product A' has the right to claim payment.

[0117] In addition, in a case in which a deposit notification of the purchase amount of the digital product A' from the purchaser (business operator β) of the digital product A' is received through the communication network **80**, in S**15**, the dealing terminal **20** (personal computer) requests the distributed ledger system **70** to register a transaction including the fact that the seller of the digital product A' does not have the right to claim payment for the digital product A'.

[0118] In addition, in a case in which the actual product A is sold, in S**19**, the dealing terminal **20** requests the distributed ledger system **70** to register a transaction including the fact that the purchaser (business operator γ) of the actual product A is the owner of the digital product A' and the location of the actual product A is registered as the location of the digital product A'. In addition, in S**20**, on the payment due date according to the sales contract with the purchaser (business operator β) of the digital product A', the dealing terminal **20** (personal computer) notifies a terminal of a financial institution via a communication network of the payment of the sales amount of the digital product A' to the account of the purchaser (business operator β) of the digital product A'.

[0119] As a result, with the digital product management support method, it is possible to support fund raising via a communication network before a business operator having a product sells the product.

[0120] (2) The dealing terminal **20** (digital product management support device) of the present embodiment includes a first registration request unit, a second registration request unit, a third registration request unit, and a processor **22** serving as a payment notification unit.

[0121] When there is a commercial dealing of the digital product A', the processor **22** requests the distributed ledger system **70** to register owner information of the digital product A' in which the purchaser of the digital product A' is a new owner of the digital product A' and a transaction indicating that the seller of the digital product A' has the right to claim payment.

[0122] In a case in which a deposit notification of the purchase amount of the digital product A' from the purchaser of the digital product A' is received via the communication network **80**, the processor **22** requests the distributed ledger system **70** to register a transaction including the fact that the seller of the digital product A' does not have the right to claim payment for the digital product A'.

[0123] In addition, in a case in which the actual product A is sold, the processor **22** requests the distributed ledger system **70** to register a transaction including the fact that the purchaser (business operator γ) of the actual product A is the owner of the digital product A' and the location of the actual product A is registered as the location of the digital product A'.

[0124] In addition, on the payment due date according to the sales contract with the purchaser of the digital product A' based on a dealing of the digital product A', the processor **22** requests the terminal of the financial institution to pay the sales amount of the digital product A' to the account of the purchaser (business operator β) of the digital product A' via the communication network **80**.

[0125] As a result, with the dealing terminal **20**, it is possible to support fund raising via a communication network before a business operator having a product sells the product.

[0126] (3) The digital product management support program of the present embodiment causes the dealing terminal **20** (personal computer) communicable with multiple terminals and the distributed ledger system **70** via the communication network **80** to implement the following functions.

[0127] First, when there is a commercial dealing of the digital product A', the distributed ledger system **70** is requested to register owner information of the digital product A' in which the purchaser (business operator β) of the digital product A' is a new owner of the digital product A' and a transaction including the fact that the seller of the digital product A' has the right to claim payment.

[0128] In addition, in a case in which a deposit notification of the purchase amount of the digital product A' from the purchaser (business operator β) of the digital product A' is received via the communication network **80**, the distributed ledger system **70** is requested to register the fact that the seller of the digital product A' does not have the right to claim payment for the digital product A'. In addition, in a case in which the actual product A is sold, the distributed ledger system **70** is requested to register a transaction including the fact that the purchaser (business operator γ) of the actual product A is the owner of the digital product A' and the location of the actual product A is registered as the location of the digital product A'.

[0129] In addition, on the payment due date according to the sales contract with the purchaser of the digital product A' based on a dealing of the digital product A', the dealing terminal **20** (processor **22**) requests the terminal of the financial institution to pay the sales amount of the digital product A' to the account of the purchaser of the digital product A' via the communication network **80**.

[0130] As a result, with the dealing terminal **20** executed by the digital product management support program, it is possible to support fund raising via a communication network before a business operator having a product sells the product.

[0131] (4) The database entity is the distributed ledger system **70**. Then, when requested to register a transaction, the distributed ledger system **70** associates a timestamp with each transaction. As a result, the distributed ledger system **70** enables determination that the transaction associated with the timestamp is not tampered.

Second Embodiment

[0132] Next, a second embodiment will be described with reference to FIGS. **5** to **7**. Since the digital product management support system **10**, the dealing terminal **20**, the terminals **30**, **40**, and **50**, the wallet management device **60**, the distributed ledger system **70**, and the communication network **80** have the same configurations as those of the first embodiment, a detailed description thereof will be omitted.

[0133] In the present embodiment, the digital product management support program stored in the storage unit **24** (see FIG. **2**) of the terminal **30** is executed by the processor **22**, whereby a transaction B**50** to be described later can be created. In addition, when the digital product management support program is executed by the processor **22**, the terminal **30** can request the distributed ledger system **70** to register the transactions B**40** and B**50** via the communication network **80**.

Operation of Second Embodiment

[0134] FIGS. **5** to **7** are sequence diagrams illustrating an outline of digital product management support of the second embodiment.

[0135] In the first embodiment, after the digital product A' is transferred to the business operator β, the business operator β is the owner of the digital product A' until the day on which the actual product A is sold.

[0136] On the other hand, in the present embodiment, a case will be described in which the digital product A' is transferred from the business operator β to the business operator β' before the day on which the actual product A is sold after the digital product A' is transferred to the business operator β. Specifically, in the sequence of FIG. **6**, a case will be described in which the digital product A' is transferred to the business operator β between T**50** (Generation of Blockchain) and S**16** (Selling of Actual Product A) in FIG. **7**. In the sequence diagrams of FIGS. **5** to **7**, the same processing as that of the first embodiment are denoted by the same reference numerals, and the description thereof will be omitted.

S30: Selling of Digital Product A'

[0137] As illustrated in FIG. **6**, when a sales contract is established between the business operator β and a business operator (business operator β') who wishes to purchase a new digital product A', digital data including an image of the digital product A' and the like, and a delivery slip and an invoice are transmitted from the terminal **30** to the terminal **50**. The delivery of the digital data including the image of the digital product A' and the like, and the delivery of the delivery slip and the invoice from the terminal **30** to the terminal **50** means the sale of the digital product A'.

[0138] The commercial dealing between the business operator β and the business operator β' may be either face-to-face dealing or dealing on a website provided by a server (not illustrated) connected to the communication network **80**.

T**80**: Creation of Transaction B**40**

[0139] As illustrated in FIG. **6**, after the sale of the digital product A' is completed, the terminal **30** creates the transaction B**40**. The creation of the transaction B**40** is performed by an input operation or the like of the input device of the terminal **30**.

[0140] The transaction B**40** includes the following (1) transaction related information, (2) digital product related information, and (3) actual product related information. Transaction B**40**

[0141] (1) Transaction related information: an identifier of a transaction, and sales contract content information between the business operators β and β' (identifiers of the business operators β and β', a dealing conclusion date and time (purchase date and time), a payment due date of a purchase amount of the digital product A' from the business operator β to the business operator β').

[0142] (2) The digital product related information includes an image and an identifier of the digital product A', owner information of the digital product A', the location *α of the digital product A', and the purchase amount of the digital product A' (trade price D': $D'=X(1-\pi')$).

[0143] X is the price of the digital product A' and the repayment amount from the vendor (business operator α) to the purchaser (business operator β) of the digital product A'. π' is the payment rate, that is, the rate from the vendor (business operator α) to the purchaser of the digital product A', and takes a positive value or a negative value depending on the market price.

[0144] In the transaction B**40**, it is indicated that the owner of the digital product A' is the business operator β'. The owner of the digital product A' corresponds to the owner information of the digital product A'.

[0145] (3) Actual product related information: the name of the actual product A associated with the digital product A', the identifier of the actual product A, the cost price of the actual product A, the sales price of the actual product A, and the location of the actual product A.

[0146] In the transaction B**10**, the location of the actual product A is the location *α of the vendor who owns the actual product A.

S**31**: Request for Registration of Transaction B**40**

[0147] As described above, when the transaction B**40** is created, the digital product management support program causes the terminal **30** to request the distributed ledger system **70** to register the transaction B**40** via the communication network **80**.

T**90**: Generation of Blockchain

[0148] In the distributed ledger system **70**, when registration of the transaction B**40** is requested, generation of a block including the transaction B**40** and generation of a blockchain are performed.

[0149] By the transaction B**40**, the location *α of the digital product A' and the fact that the owner of the digital product A' is the business operator β' are registered in the database of the distributed ledger system **70**.

S**32**: Payment Instruction for Amount

[0150] In S**32**, the terminal **50** of the business operator β' instructs the wallet management device **60** to pay the purchase amount (trade price D') of the digital product A' to the vendor (business operator β) via the communication network **80**. In response to the payment instruction, the wallet management device **60** remits money from the account of the business operator who has given the payment instruction to another wallet management device that manages the account of the vendor (business operator β).

S**33**: Notification of Deposit of Amount

[0151] In S**33**, a deposit notification (for example, push notification) of the amount from the business operator β' is received via the other wallet management device to the account of the vendor via the communication network **80**. The currency to be deposited in the account is a stablecoin or cash.

T**100**: Creation of Transaction B**50**

[0152] As illustrated in FIG. **6**, when a notification of deposit of the amount from the business operator β' to the account of the vendor via the communication network **80**, the digital product management support program causes the terminal **30** to create the transaction B**50**.

Transaction B**50**

[0153] In the transaction B**50**, the history information indicating that the amount has been deposited from the business operator β' is created, and (1) the transaction related information, (2) the digital product related information, and (3) the actual product related information are created. Since (1) the transaction related information, (2) the digital product related information, and (3) the actual product related information of the transaction B**50** are similar to those of the transaction B**40**, the description thereof will be omitted.

S**34**: Request for Registration of Transaction B**50**

[0154] As described above, when the transaction B**50** is created, the digital product management support program causes the terminal **30** to request the distributed ledger system **70** to register the transaction B**50** via the communication network **80**.

T**110**: Generation of Blockchain

[0155] As illustrated in FIG. **7**, in the distributed ledger system **70**, when registration of the transaction B**50** is requested, generation of a block including the transaction B**50** and generation of a blockchain are performed.

[0156] By the transaction B**50**, the history information indicating that the amount has been deposited from the business operator β', the location *α of the digital product A', and the fact that the owner of the digital product A' is the business operator β' are registered in the database of the distributed ledger system **70**.

[0157] Hereinafter, in the sequence of FIG. **7**, since the processing of S**16** to S**21**, T**60**, and T**70** is similar to that of the first embodiment, the description thereof will be omitted.

[0158] The second embodiment has the following effects.

[0159] (1) Also in the second embodiment, the same effects as those of the first embodiment are obtained.

[0160] With the digital product management support method, when the business operator β is a financial company or a financial intermediary, a product of the business operator α is fluidized to perform financial commercialization such as securitization. As a result, when the business operator β' is a general investor, the business operator β can sell the digital product A' that has been financially commercialized to the general investor.

[0161] In addition, the business operator β', who is a general investor, waits for the fund collection from the business operator γ who is an end user, or becomes a member of a group of the business operators γ who are end users, and purchases the actual product A, whereby a dealing of the financial product can be completed.

[0162] The business operator γ serving as the end user can also take over the amount scheduled to be paid to the business operator α as liabilities of the business operator γ. In this case, the obligation to fulfil the debt of the business operator α (the obligation to complete the sale of the actual product A and the collection obligation) is transferred to the business operator γ. In such a case, it is sufficient that the business operator β approves. However, since this act is based on the approval of the business operator β, when the business operator γ does not have the ability to pay, the business operator γ does not defraud.

Modifications of First Embodiment and Second Embodiment

[0163] Although not illustrated, modifications of the first embodiment and the second embodiment in a case in which the location of the actual product A is changed before S**11** (selling of the digital product A') will be described.

[0164] Each time there is a change in the location of the actual product A before S**11** (selling of the digital product A') in FIGS. **3** and **5**, a new transaction B**0** is created in the dealing terminal **20**. The new transaction B**0** includes digital product related information including the changed location of the digital product A', actual product related information including the changed location of the actual product A, and transaction related information.

[0165] Then, the digital product management support program causes the dealing terminal **20** to request the distributed ledger system **70** to register the new transaction B**0** via the communication network **80**. In response to the registration request of the new transaction B**0**, the distributed ledger system **70** generates a blockchain.

[0166] As a result, the new location *α of the digital product A' and the fact that the owner of the digital product A' is the vendor (business operator α) of the digital product A' are registered in the blockchain.

Third Embodiment

[0167] Next, a third embodiment will be described with reference to FIGS. **8** to **11**.

2.1. Digital Product Management Support System **10**

[0168] As illustrated in FIG. **8**, the digital product management support system **10** of the present embodiment includes the dealing terminal **20**, the terminals **30**, **40**, and **50**, the wallet management device **60**, a database management server **90**, and a time stamping authority **100**. Although three terminals **30**, **40**, and **50** are shown for illustrative purposes, four or more terminals may be provided.

[0169] The storage unit **24** (see FIG. **2**) of the dealing terminal **20** stores a digital product management support program, an application program such as a web browser, and various data acquired by execution processing of the digital product management support program.

[0170] The dealing terminal **20**, the terminals **30**, **40**, and **50**, the wallet management device **60**, the database management server **90**, and the time stamping authority **100** are connected via the communication network **80**. The communication network **80** includes, for example, the Internet, a WAN, and the like.

2.2. Dealing Terminal **20**, Terminals **30**, **40**, **50**

[0171] Since the hardware configuration of the dealing terminal **20** and the terminals **30**, **40**, and **50** is similar to that of the first embodiment, the description thereof will be omitted. Business operators who use the dealing terminal **20** and the terminals **30**, **40**, and **50** are both users who are qualified to access the database management server **90**. These business operators can confirm the content of transactions in a digital product management database stored in the database management server **90** by inputting the respective identifiers and passwords of the business operators using the input devices of the dealing terminal **20** and the terminals **30**, **40**, and **50**. The password is preferably a public key cryptosystem.

[0172] The dealing terminal **20** of the present embodiment corresponds to the digital product management support device, and a personal computer constituting the dealing terminal **20** corresponds to the computer of the present disclosure.

2.3. Database Management Server **90**

[0173] The database management server **90** stores the digital product management database including a history of commercial dealings of the digital product A'.

[0174] The digital product management database is a database that manages a history related to a commercial dealing for each digital product. The database is time series data in which transactions of the commercial dealings of the digital product A' is arranged according to timestamps. That is, a transaction is associated with a timestamp.

[0175] The database management server **90** corresponds to a database entity.

2.4. Time Stamping Authority **100**

[0176] The time stamping authority **100** is an organization certified by the Minister for Internal Affairs and Communications for time stamping operations. In the present embodiment, it is possible to obtain high reliability with respect to the data related to the dealing of the digital product A' by authenticating the data related to the dealing of the digital product by adding a timestamp of the time stamping authority **100**.

2.5. Wallet Management Device **60**

[0177] Since the wallet management device **60** is similar to that of the first embodiment, the description thereof will be omitted.

Operation of Third Embodiment

[0178] Next, operation of the digital product management support system **10**, the dealing terminal **20**, and the digital product management support program of the present embodiment will be described with reference to FIGS. **9** to **11**. FIGS. **9** to **11** are sequence diagrams illustrating an outline of digital product management support.

T**0**: Creation of Transaction B**0**

[0179] As illustrated in FIG. **9**, the dealing terminal **20** of the vendor (business operator α) creates a transaction B**0**. Since the creation of the transaction B**0** and the contents of the transaction B**0** are similar to those of the first embodiment, the description thereof will be omitted.

S**100**: Request for Timestamp, and S**110**: Return of Timestamp

[0180] In S**100**, the dealing terminal **20** requests the time stamping authority **100** to add a timestamp to the transaction B**0** attached with an electronic signature. This request is made by an operation of the input device of the dealing terminal **20**.

[0181] In S**110**, the time stamping authority **100** returns the transaction B**0** associated with the timestamp to the dealing terminal **20**.

S**120**: Request for Registration of Transaction B**0**

[0182] As described above, when the transaction B**0** with the timestamp is returned, the digital product management support program causes the dealing terminal **20** to request the database management server **90** to register the transaction B**0** via the communication network **80**.

T**210**: Registration of Default Data of Digital Product A'

[0183] When there is a registration request of the transaction B**0**, the database management server **90** registers default data of the digital product A' in the digital product management database.

[0184] As a result, the location *α of the digital product A' and the fact that the owner of the digital product A' is the vendor (business operator α) of the digital product A' are registered in the digital product management database of the database management server **90**.

S**130**: Selling of Digital Product A'

[0185] As illustrated in FIG. **9**, when a sales contract is established between the vendor (business operator α) and a business operator (business operator β) who wishes to purchase the digital product A', digital data including an image of the digital product A' and the like, and a delivery slip and an invoice are transmitted from the dealing terminal **20** to the terminal **30**. This transmission is executed by the processor **22** of the dealing terminal **20** when an input device connected to the input/output interface **26** is operated. The delivery of the digital data including the image of the digital product A' and the like, and the delivery of the delivery slip and the invoice from the dealing terminal **20** to the terminal **30** means the sale of the digital product A'.

[0186] The commercial dealing between the business operator α and the business operator β may be either face-to-face dealing or dealing on a website provided by a server (not illustrated) connected to the communication network **80**.

T**20**: Creation of Transaction B**10**

[0187] As illustrated in FIG. **9**, the dealing terminal **20** of the vendor (business operator α) creates a transaction B**10**. Since the creation of the transaction B**10** and the contents of the transaction B**10** are similar to those of the first embodiment, the description thereof will be omitted.

S**140**: Request for Timestamp, and S**150**: Return of Timestamp

[0188] In S**140**, the dealing terminal **20** requests the time stamping authority **100** to add a timestamp to the transaction B**10** attached with an electronic signature. This request is made by an operation of the input device of the dealing terminal **20**.

[0189] In S**150**, the time stamping authority **100** returns the transaction B**10** associated with the timestamp to the dealing terminal **20**.

S**160**: Request for Registration of Transaction B**10**

[0190] As described above, when the transaction B**10** with the timestamp is returned, the digital product management support program causes the dealing terminal **20** to request the database management server **90** to register the transaction B**10** via the communication network **80**.

[0191] The registration request includes a registration request of the owner information of the digital product A' in which the purchaser of the digital product A' is the new owner of the digital product A', and a registration request indicating that the seller of the digital product A' has the right to claim payment for the digital product A'.

T**220**: Additional Registration of Data of Digital Product A'

[0192] As illustrated in FIG. **10**, when there is a registration request of the transaction B**10**, the database management server **90** additionally registers data of the digital product A' in the digital product management database.

[0193] By the transaction B**10**, the location *α of the digital product A', information indicating that the owner of the digital product A' is the purchaser (business operator β) of the digital product A', and that the seller of the digital product A' has the right to claim payment for the digital product A' are registered in the digital product management database of the database management server **90**.

S**170**: Payment Instruction for Amount

[0194] In S**170**, the terminal **30** of the business operator β instructs the wallet management device **60** to pay the purchase amount (trade price D) of the digital product A' to the vendor (business operator α) via the communication network **80**. In response to the payment instruction, the wallet management device **60** remits money from the account of the business operator who has given the payment instruction to another wallet management device that manages the account of the vendor (business operator α).

S**180**: Notification of Deposit of Amount

[0195] In S**180**, a deposit notification (for example, push notification) of the amount from the business operator β is received via the other wallet management device to the account of the vendor via the communication network **80**. The currency to be deposited in the account is a stablecoin or cash.

T**40**: Creation of Transaction B**20**

[0196] As illustrated in FIG. **10**, when a notification of deposit of the amount from the business operator β to the account of the vendor is received via the communication network **80**, the digital product management support program causes the dealing terminal **20** to create a transaction B**20**.

Transaction B**20**

[0197] Since the creation and the contents of the transaction B**20** are similar to those of the first embodiment, the description thereof will be omitted.

[0198] Therefore, in the transaction B**20**, similarly to the transaction B**10**, the owner of the digital product A' is the purchaser (business operator β), and there is no change from the transaction B**10**. In the transaction B**20**, the location *α of the digital product A' is not changed from the transaction B**10**.

[0199] The location *α of the digital product A' corresponds to the location information of the digital product A'.

S**190**: Request for Timestamp, and S**200**: Return of Timestamp

[0200] In S**190**, the dealing terminal **20** requests the time stamping authority **100** to add a timestamp to the transaction B**20** attached with an electronic signature. This request is made by an operation of the input device of the dealing terminal **20**.

[0201] In S**200**, the time stamping authority **100** returns the transaction B**20** associated with the timestamp to the dealing terminal **20**.

S**210**: Request for Registration of Transaction B**20**

[0202] As described above, when the transaction B**20** with the timestamp is returned, the digital product management support program causes the dealing terminal **20** to request the database management server **90** to register the transaction B**20** via the communication network **80**.

[0203] The registration request includes a registration request indicating that the seller of the digital product A' does not have a right to claim payment for the digital product A'.

T**230**: Additional Registration of Data of Digital Product A'

[0204] As illustrated in FIG. **11**, when there is a registration request of the transaction B**20**, the database management server **90** additionally registers data of the digital product A' in the digital product management database.

[0205] As a result, the location *α of the digital product A', information indicating that the owner of the digital product A' is the purchaser (business operator β), and that the seller of the digital product A' does not have the right to claim payment for the digital product A' are registered in the digital product management database of the database management server **90**.

S**220**, S**230**, S**240**, and T**60**

[0206] S**220**, S**230**, S**240**, and T**60** illustrated in FIG. **11** are the same as S**16** (selling of the actual product A), S**17** (payment instruction for the purchase amount of the actual product A), S**18** (notification of deposit of the purchase amount), and T**60** (creation of the transaction B**30**) in the first embodiment, respectively. Therefore, the description of S**220**, S**230**, S**240**, and T**60** is omitted.

[0207] S**250**: Request for Timestamp, and S**260**: Return of Timestamp

[0208] In S**250** after T**60** illustrated in FIG. **11**, the dealing terminal **20** requests the time stamping authority **100** to add a timestamp to the transaction B**30** attached with an electronic signature. This request is made by an operation of the input device of the dealing terminal **20**.

[0209] In S**260**, the time stamping authority **100** returns the transaction B**30** associated with the timestamp to the dealing terminal **20**.

S**270**: Request for Registration of Transaction B**30**

[0210] When the transaction B**30** with the timestamp is returned, the digital product management support program causes the dealing terminal **20** to request the database management server **90** to register the transaction B**30** via the communication network **80**.

[0211] The registration request includes a registration request of owner information of the digital product A' in which the purchaser (business operator γ) of the actual product A is a new owner of the digital product A', and a registration request of a location of the digital product A'.

T**240**: Additional Registration of Data of Digital Product A'

[0212] When there is a registration request of the transaction B**30**, the database management server **90** additionally registers data of the digital product A' in the digital product management database.

[0213] As a result, the location *γ of the digital product A' and the fact that the owner of the digital product A' is the purchaser (business operator γ) of the actual product A are registered in the database of the database management server **90**.

S**280** and S**290**

[0214] S**280** and S**290** illustrated in FIG. **11** are the same as S**20** (payment instruction for the repayment amount on the repayment due date (payment due date) to the business operator β), and S**21** (notification of deposit of the repayment amount) in the first embodiment, respectively. Therefore, the description of S**280** and S**290** is omitted.

[0215] The third embodiment has the following features.

[0216] (1) The digital product management support system **10** and the digital product management support system **10** according to the present embodiment request the database management server **90** to register a transaction associated with a timestamp by the time stamping authority **100**. Then, the database management server **90** registers the transaction associated with the timestamp in the digital product management database. As a result, since the timestamp is associated with the transaction registered in the digital product management database of the database management server **90**, there is no tampering.

Fourth Embodiment

[0217] Next, a fourth embodiment will be described with reference to FIGS. **12** to **15**. Since the digital product management support system **10**, the dealing terminal **20**, the terminals **30**, **40**, and **50**, the wallet management device **60**, the database management server **90**, and the communication network **80** have the same configurations as those of the third embodiment, a detailed description thereof will be omitted.

[0218] In the present embodiment, the digital product management support program stored in the storage unit **24** (see FIG. **2**) of the terminal **30** is executed by the processor **22**, whereby a transaction T**270** to be described later can be created. In addition, when the digital product management support program is executed by the processor **22**, the terminal **30** can request the database management server **90** to register the transactions T**250** and T**270** via the communication network **80**.

Operation of Fourth Embodiment

[0219] Next, operation of the digital product management support system **10**, the dealing terminal **20**, and the digital product management support program of the present embodiment will be described with reference to FIGS. **12** to **15**. FIGS. **12** to **15** are sequence diagrams illustrating an outline of digital product management support.

[0220] Since the following part from T**0** (creation of the transaction B**0**) in FIGS. **12** to T**230** (additional registration of data of the digital product A') in FIG. **13** is the same processing as that of the third embodiment, the description thereof will be omitted.

[0221] T**0**: Creation of Transaction B**0**

[0222] S**100**: Request for Timestamp, and S**110**: Return of Timestamp

[0223] S**120**: Request for Registration of Transaction B**0**

[0224] T**210**: Registration of Default Data of Digital Product A'

[0225] S**130**: Selling of Digital Product A'

[0226] T**20**: Creation of Transaction B**10**

[0227] S**140**: Request for Timestamp, and S**150**: Return of Timestamp

[0228] S**160**: Request for Registration of Transaction B**10**

[0229] T**220**: Additional Registration of Data of Digital Product A'

[0230] S**170**: Payment Instruction for Amount

[0231] S**180**: Notification of Deposit of Amount

[0232] T**40**: Creation of Transaction B**20**

[0233] S**190**: Request for Timestamp, and S**200**: Return of Timestamp

[0234] S**210**: Request for Registration of Transaction B**20**

[0235] T**230**: Additional Registration of Data of Digital Product A'

[0236] S**300**: Selling of Digital Product A'

[0237] As illustrated in FIG. **13**, when a sales contract is established between the business operator β and a business operator (business operator β') who wishes to purchase a new digital product A', digital data including an image of the digital product A' and the like, and a delivery slip and an invoice are transmitted from the terminal **30** to the terminal **50**. The delivery of the digital data including the image of the digital product A' and the like, and the delivery of the delivery slip and the invoice from the terminal **30** to the terminal **50** means the sale of the digital product A'.

[0238] The commercial dealing between the business operator β and the business operator β' may be either face-to-face dealing or dealing on a website provided by a server (not illustrated) connected to the communication network **80**.

T**250**: Creation of Transaction B**40**

[0239] As illustrated in FIG. **13**, after the sale of the digital product A' is completed, the terminal **30** creates the transaction B**40**. The creation of the transaction B**40** is performed by an input operation or the like of the input device of the terminal **30**.

[0240] Since the contents of the transaction B**40** are similar to those of the transaction B**40** of the second embodiment, the description thereof will be omitted.

S**301**: Request for Timestamp, and S**302**: Return of Timestamp

[0241] In S**301**, the terminal **30** requests the time stamping authority **100** to add a timestamp to the transaction B**40** attached with an electronic signature. This request is made by an operation of the input device of the terminal **30**.

[0242] In S**302**, the time stamping authority **100** returns the transaction B**40** associated with the timestamp to the terminal **30**.

S**303**: Request for Registration of Transaction B**40**

[0243] As described above, when the transaction B**40** with the timestamp is returned, the digital product management support program causes the terminal **30** to request the database management server **90** to register the transaction B**40** via the communication network **80**.

T**260**: Additional Registration of Data of Digital Product A'

[0244] When there is a registration request of the transaction B**40**, the database management server **90** additionally registers data of the digital product A' in the digital product management database.

[0245] As a result, the location *α of the digital product A' and the information indicating that the owner of the digital product A' is the purchaser (business operator β') of the digital product A' are registered in the digital product management database of the database management server **90**.

S**304**: Payment Instruction for Amount

[0246] In S**304**, the terminal **50** of the business operator β' instructs the wallet management device **60** to pay the purchase amount (trade price D') of the digital product A' to the vendor (business operator β) via the communication network **80**. In response to the payment instruction, the wallet management device **60** remits money from the account of the business operator who has given the payment instruction to another wallet management device that manages the account of the vendor (business operator β).

S**305**: Notification of Deposit of Amount

[0247] In S**305**, a deposit notification (for example, push notification) of the amount from the business operator β' is received via the other wallet management device to the account of the vendor via the communication network **80**. The currency to be deposited in the account is a stablecoin or cash.

T**270**: Creation of Transaction B**50**

[0248] As illustrated in FIG. **14**, when a notification of deposit of the amount from the business operator β' to the account of the vendor via the communication network **80**, the digital product management support program causes the terminal **30** to create the transaction B**50**.

[0249] Since the contents of the transaction B**50** are similar to those of the transaction B**50** of the second embodiment, the description thereof will be omitted.

S**306**: Request for Timestamp, and S**307**: Return of Timestamp

[0250] In S**306**, the terminal **30** requests the time stamping authority **100** to add a timestamp to the transaction B**50** attached with an electronic signature. This request is made by an operation of the input device of the terminal **30**.

[0251] In S**307**, the time stamping authority **100** returns the transaction B**50** associated with the timestamp to the terminal **30**.

S**308**: Request for Registration of Transaction B**50**

[0252] As described above, when the transaction B**50** with the timestamp is returned, the digital product management support program causes the terminal **30** to request the database management server **90** to register the transaction B**50** via the communication network **80**.

T**280**: Additional Registration of Data of Digital Product A'

[0253] As illustrated in FIG. **15**, when there is a registration request of the transaction B**50**, the database management server **90** additionally registers data of the digital product A' in the digital product management database.

[0254] As a result, the history information indicating that the amount has been deposited from the business operator β', the location *α of the digital product A' and the information indicating that the owner of the digital product A' is the purchaser (business operator β') of the digital product A' are registered in the digital product management database of the database management server **90**.

S**220** to S**290** and T**240**: Additional Registration of Data of Digital Product A'

[0255] S**220** to S**290** and T**240** (additional registration of data of the digital product A') in FIG. **15** are the same as S**220** to S**290** and T**240** (additional registration of data of the digital product A') in the third embodiment, respectively. Therefore, the description thereof is omitted.

[0256] The fourth embodiment has the same effects as those of the third embodiment.

Modifications of Third Embodiment and Fourth Embodiment

[0257] Although not illustrated, modifications of the third embodiment and the fourth embodiment in a case in which the location of the actual product A is changed before S**130** (selling of the digital product A').

[0258] Each time there is a change in the location of the actual product A before S**130** (selling of the digital product A') in FIGS. **9** and **12**, a new transaction B**0** is created in the dealing terminal **20**. The new transaction B**0** includes digital product related information including the changed location of the digital product A', actual product related information including the changed location of the actual product A, and transaction related information.

[0259] Then, the dealing terminal **20** requests the time stamping authority **100** to add a timestamp to the new transaction B**0** attached with an electronic signature. This request is made by an operation of the input device of the dealing terminal **20**.

[0260] The time stamping authority **100** returns the new transaction B**0** associated with the timestamp to the dealing terminal **20**.

[0261] When the transaction B**0** with the timestamp is returned, the digital product management support program causes the dealing terminal **20** to request the database management server **90** to register the transaction B**0** via the communication network **80**.

[0262] When there is a registration request of the new transaction B**0**, the database management server **90** registers default data of the digital product A' in the digital product management database.

[0263] As a result, the new location *α of the digital product A' and the fact that the owner of the digital product A' is the vendor (business operator α) of the digital product A' are registered in the digital product management database of the database management server **90**.

[0264] The present embodiment can be modified and implemented as follows.

[0265] The present embodiment and the following modifications can be implemented in combination with each other within a range not technically contradictory.

[0266] Although the blockchain is adopted in the distributed ledger system **70** of the first embodiment and the third embodiment, the disclosure is not limited thereto. Instead of the blockchain, a directed acyclic graph (DAG) may be adopted.

[0267] In the DAG, when a dealer creates a new transaction in association with a previously published unapproved transaction, the dealer approves the previously published unapproved transaction. Therefore, the transaction created by the dealer will also be approved by the fact that there is an unapproved transaction to be published later.

[0268] In the DAG, when an unapproved transaction equal to or greater than a threshold is directly or indirectly approved, a new transaction created by the user is considered to be agreed in the network by a consensus algorithm.

[0269] The DAG differs from the blockchain in the consensus algorithm, but the same transaction configuration can be adopted. That is, the processing using the blockchain as the distributed ledger system and the processing using the DAG can be performed using the same transaction.

[0270] In the second embodiment and the fourth embodiment, the digital product A' is transferred from the business operator β to a single business β' by purchase, but the business β' is not limited to a single business operator and may be multiple business operators.

[0271] In the embodiments, the deposit to an account is made in a stablecoin or cash, but the deposit is not limited thereto, and may be made in credit or point.

[0272] In the modifications of the embodiments, the case in which there is a change in the location of the actual product A has been described. In a case in which the owner himself/herself is changed, the changed owner may update and register the changed owner information in the distributed ledger system **70** or the database management server **90**.

[0273] In the embodiment, the case in which the actual product A is a single product has been described. However, even if the actual product A is multiple actual products, if each of the actual products A is individually managed and sold to multiple business operators γ, it is similar to the case of a single product.

[0274] When there are multiple actual products A to be sold by the business operator α and the actual products A to be sold are not sold out to the business operator γ, the business operator α may buy back the sold digital products A' from the business operator β based on the sales contract between the business operators α and β. Alternatively, the business operator β may continue to store the digital product A' and continue to entrust sales of the multiple actual products A.

[0275] In addition, the embodiments are not limited to the case in which the number of the actual products A is singular, and the digital product A' associated with the actual product A is a single product. In a case in which the number of the actual product A is a singular and there are multiple digital products A' associated with the actual product A, it is sufficient that the total amount of the multiple digital products A' is the same as the amount of the actual product A.

[0276] Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

What is claimed is:

1. A digital product management support method for causing a computer of a dealing terminal to execute the following steps upon dealing of a digital product associated with an actual product, the method comprising:

when there is a commercial dealing of the digital product, requesting, by the computer, a database entity under a communication network to register a transaction including owner information in which a purchaser of the digital product is a new owner of the digital product and a fact that a seller of the digital product has a right to claim payment for the digital product;

in a case in which a deposit notification of a purchase amount of the digital product from the purchaser of the digital product is received via the communication network, requesting, by the computer, the database entity to register a transaction including a fact that the seller of the digital product does not have the right to claim payment;

in a case in which the actual product is sold, requesting the database entity to register a transaction including a fact that the purchaser of the actual product is an owner of the digital product and a location of the actual product is registered as a location of the digital product; and

on a payment due date according to a sales contract with the purchaser of the digital product based on the dealing of the digital product, notifying, by the computer, a terminal of a financial institution of payment of a sales amount of the digital product to an account of the purchaser of the digital product via the communication network.

2. The digital product management support method according to claim **1**, wherein

the database entity is a distributed ledger system in which multiple node computers are connected to each other wirelessly or by wire via a P2P network and share a blockchain, and

the database entity is configured to associate a timestamp with the transaction when there is a registration request of the transaction.

3. The digital product management support method according to claim **1**, wherein

the transaction is associated with a timestamp, and

the database entity is a database management server.

4. A digital product management support device communicable with multiple terminals and a database entity via a communication network and configured to support management of dealing of a digital product associated with an actual product, the device comprising:

a first registration request unit including a processor configured to, when there is a commercial dealing of the digital product, request the database entity under the communication network to register a transaction including owner information in which a purchaser of the digital product is a new owner of the digital product and a fact that a seller of the digital product has a right to claim payment for the digital product;

a second registration request unit including a processor configured to, in a case in which a deposit notification of a purchase amount of the digital product from the purchaser of the digital product is received via the communication network, request the database entity to register a transaction including a fact that the seller of the digital product does not have the right to claim payment;

a third registration request unit including a processor configured to, in a case in which the actual product is sold, request the database entity to register a transaction including a fact that the purchaser of the actual product is an owner of the digital product and a location of the actual product is registered as a location of the digital product; and

a payment notification unit including a processor configured to request a terminal of a financial institution to pay a sales amount of the digital product to an account of the purchaser of the digital product via the communication network on a payment due date according to a sales contract with the purchaser of the digital product based on the dealing of the digital product.

5. A computer-readable medium recording a digital product management support program implemented by multiple terminals and a computer communicable with a database entity via a communication network, the digital product management support program including an instruction that causes the computer to:

when there is a commercial dealing of a digital product associated with an actual product, request the database entity under the communication network to register a transaction including owner information in which a purchaser of the digital product is a new owner of the digital product and a fact that a seller of the digital product has a right to claim payment for the digital product;

in a case in which a deposit notification of a purchase amount of the digital product from the purchaser of the digital product is received via the communication network, request the database entity to register a transaction including a fact that the seller of the digital product does not have the right to claim payment;

in a case in which the actual product is sold, request the database entity to register a transaction including a fact that the purchaser of the actual product is an owner of the digital product and a location of the actual product is registered as a location of the digital product; and

notify a terminal of a financial institution of payment of a sales amount of the digital product to an account of the purchaser of the digital product via the communication network on a payment due date according to a sales contract with the purchaser of the digital product based on the dealing of the digital product.

6. A digital product management support system, comprising a digital product management support device, multiple terminals, and a database entity that are connected via a communication network, the digital product management support device including:

a first registration request unit including a processor configured to, when there is a commercial dealing of a digital product associated with an actual product, request the database entity under the communication network to register a transaction including owner information in which a purchaser of the digital product is a new owner of the digital product and a fact that a seller of the digital product has a right to claim payment for the digital product;

a second registration request unit including a processor configured to, in a case in which a deposit notification of a purchase amount of the digital product from the purchaser of the digital product is received via the communication network, request the database entity to register a transaction including a fact that the seller of the digital product does not have the right to claim payment;

a third registration request unit including a processor configured to, in a case in which the actual product is sold, request the database entity to register a transaction including a fact that the purchaser of the actual product is an owner of the digital product and a location of the actual product is registered as a location of the digital product; and

a payment notification unit including a processor configured to request a terminal of a financial institution to pay a sales amount of the digital product to an account of the purchaser of the digital product via the communication network on a payment due date according to a sales contract with the purchaser of the digital product based on the dealing of the digital product,

wherein the database entity is configured to perform processing according to a registration request of the first registration request unit, a registration request of the second registration request unit, and a registration request of the third registration request unit.

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